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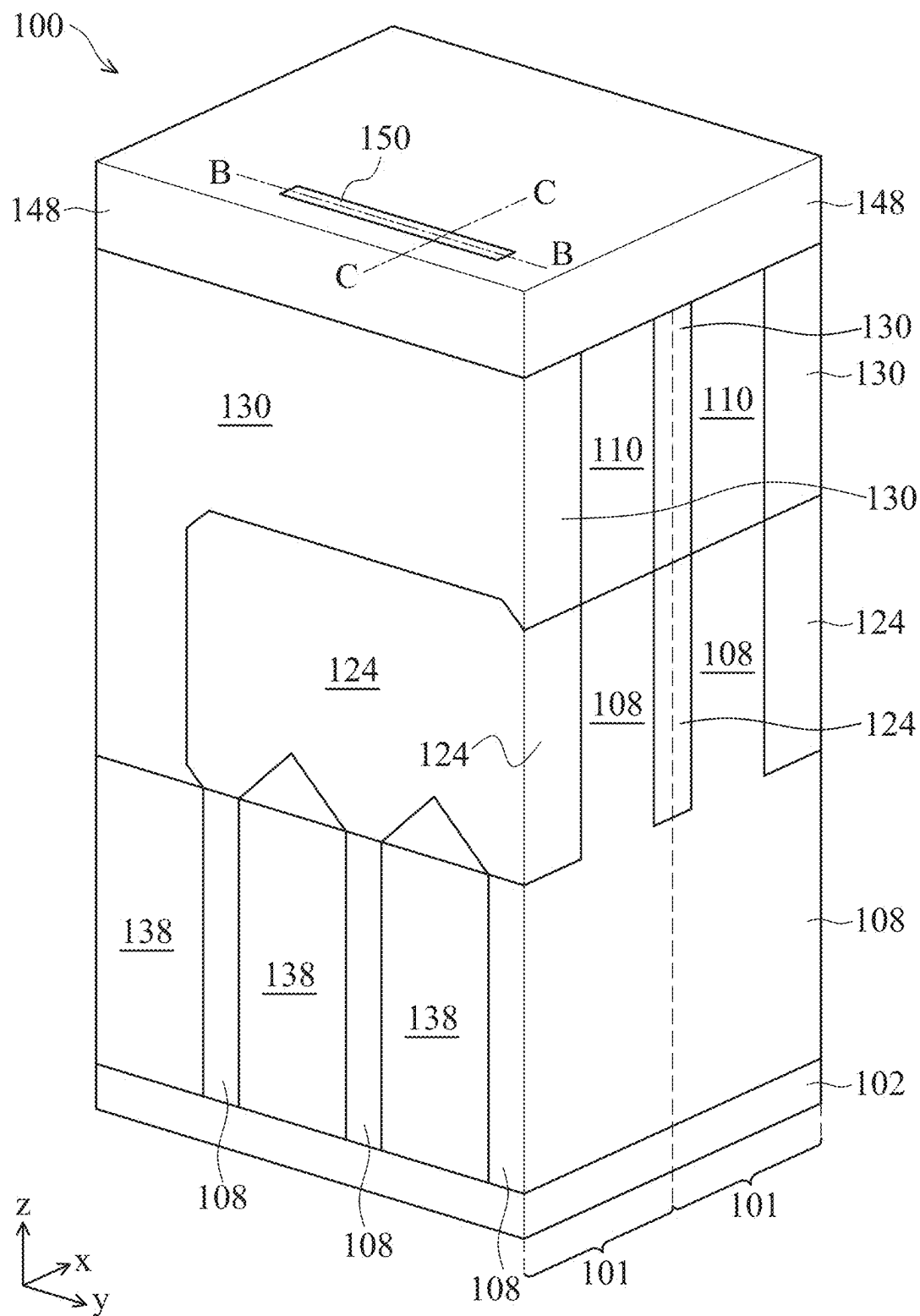


Fig. 1A

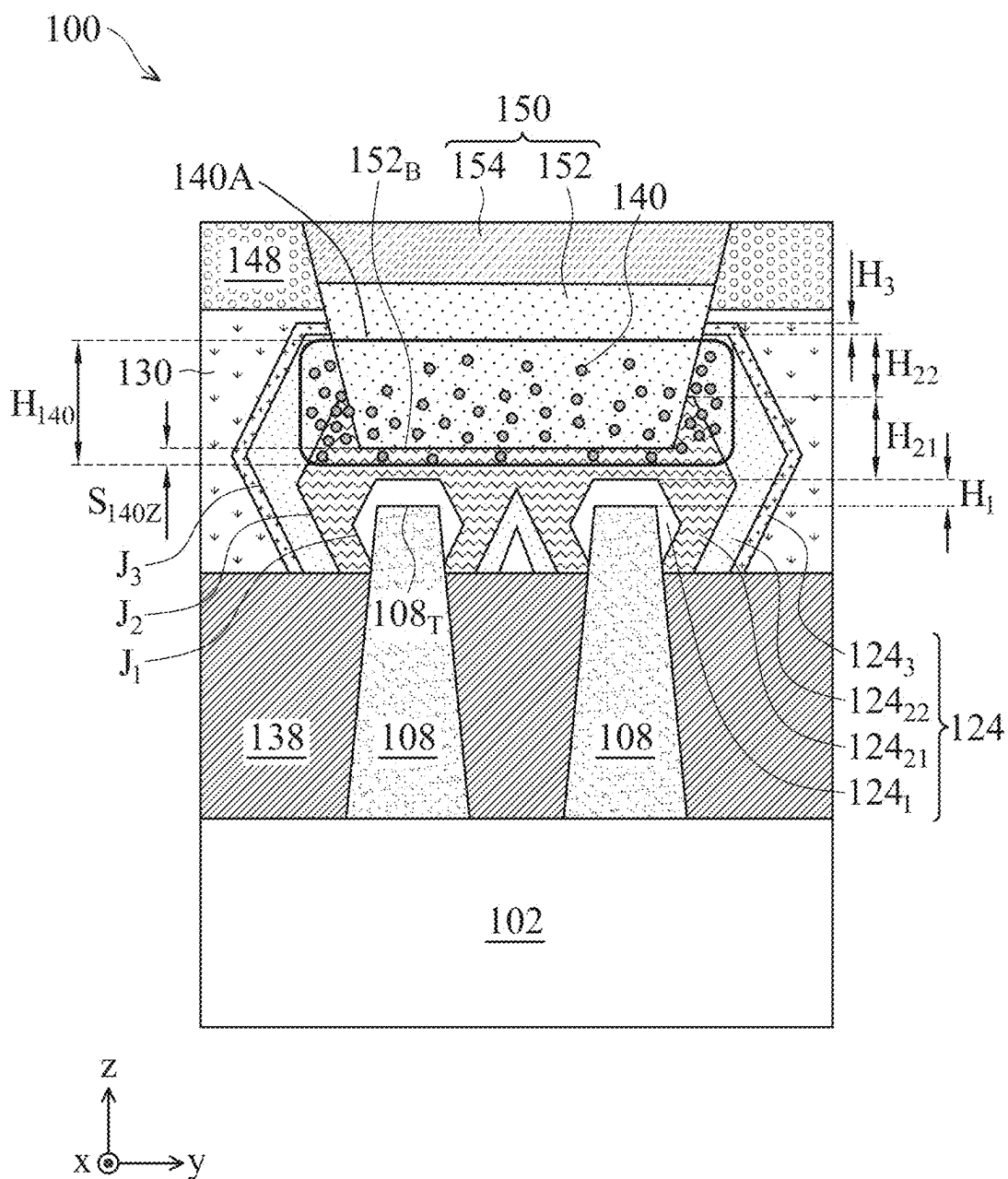


Fig. 1B

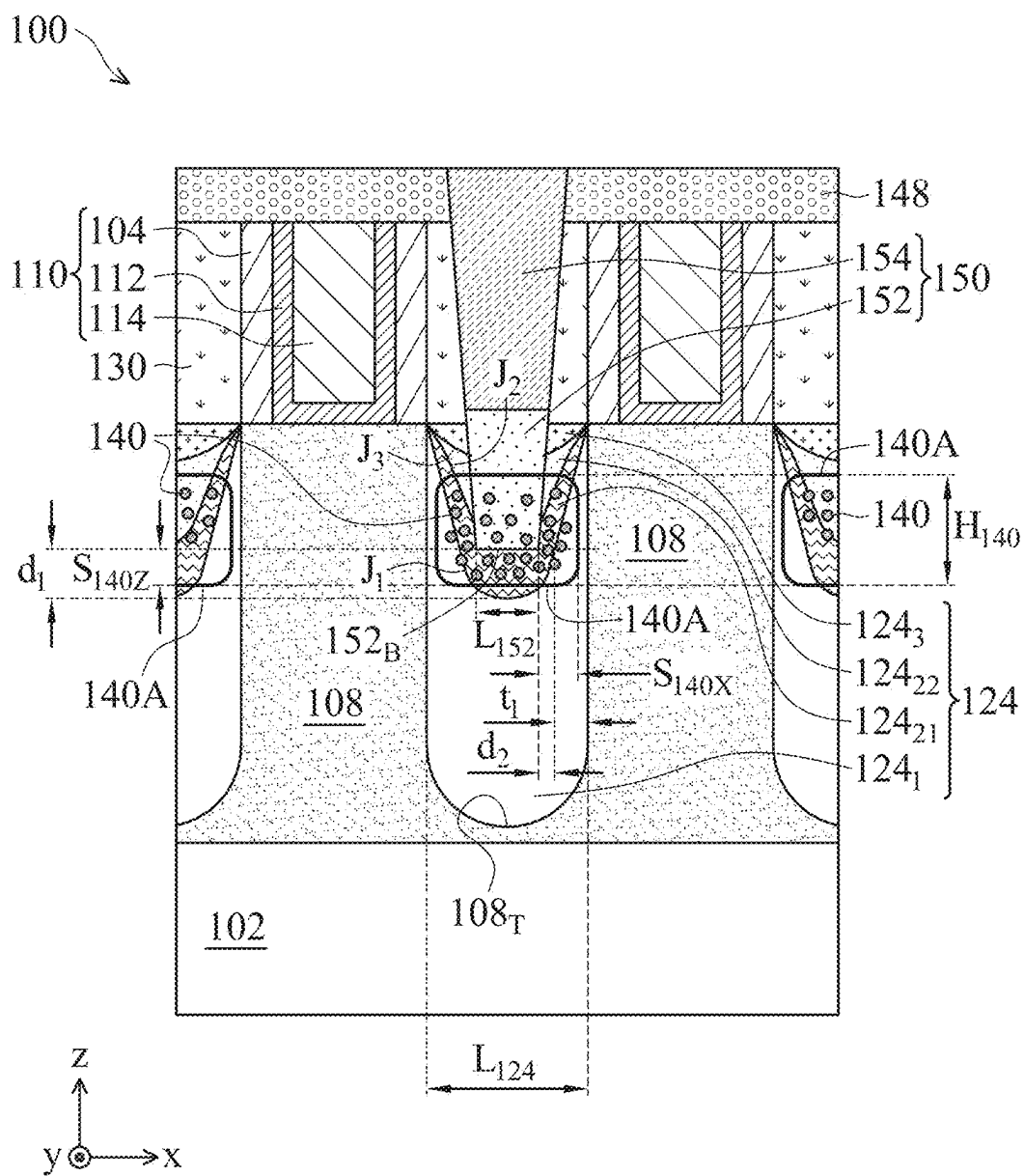


Fig. 1C

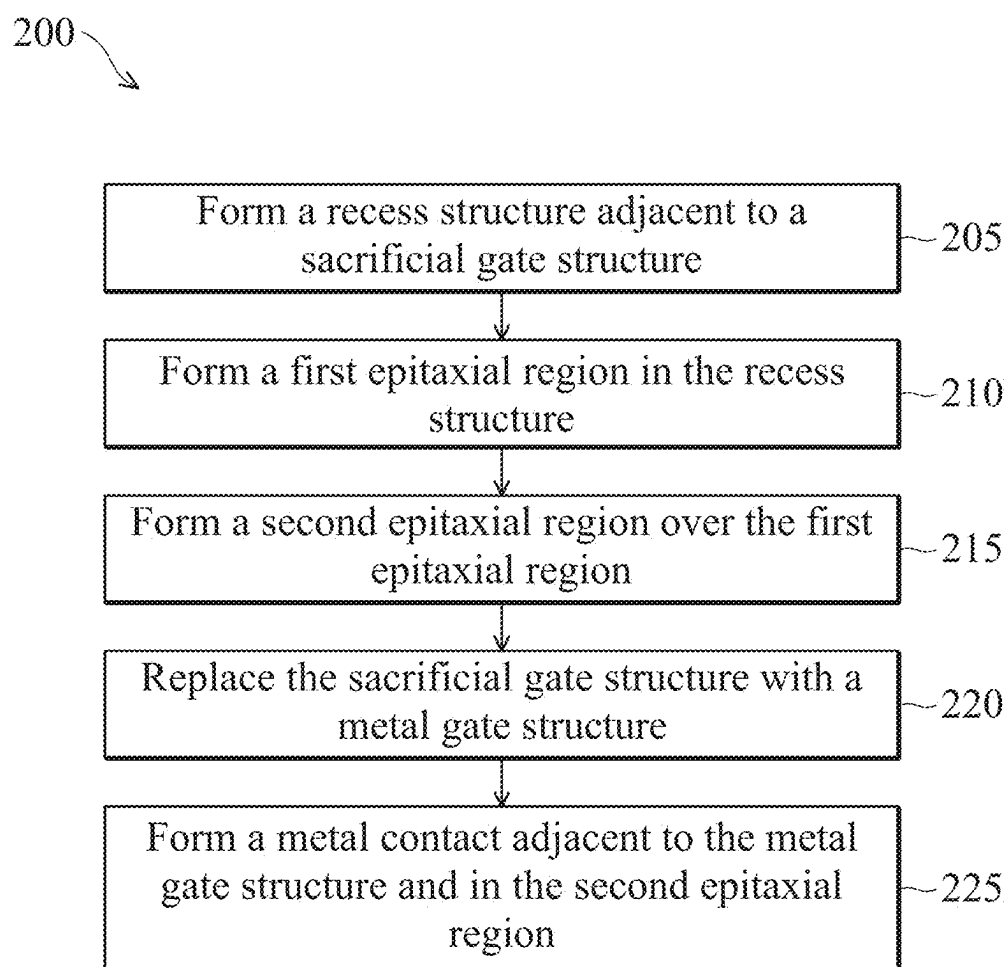


Fig. 2

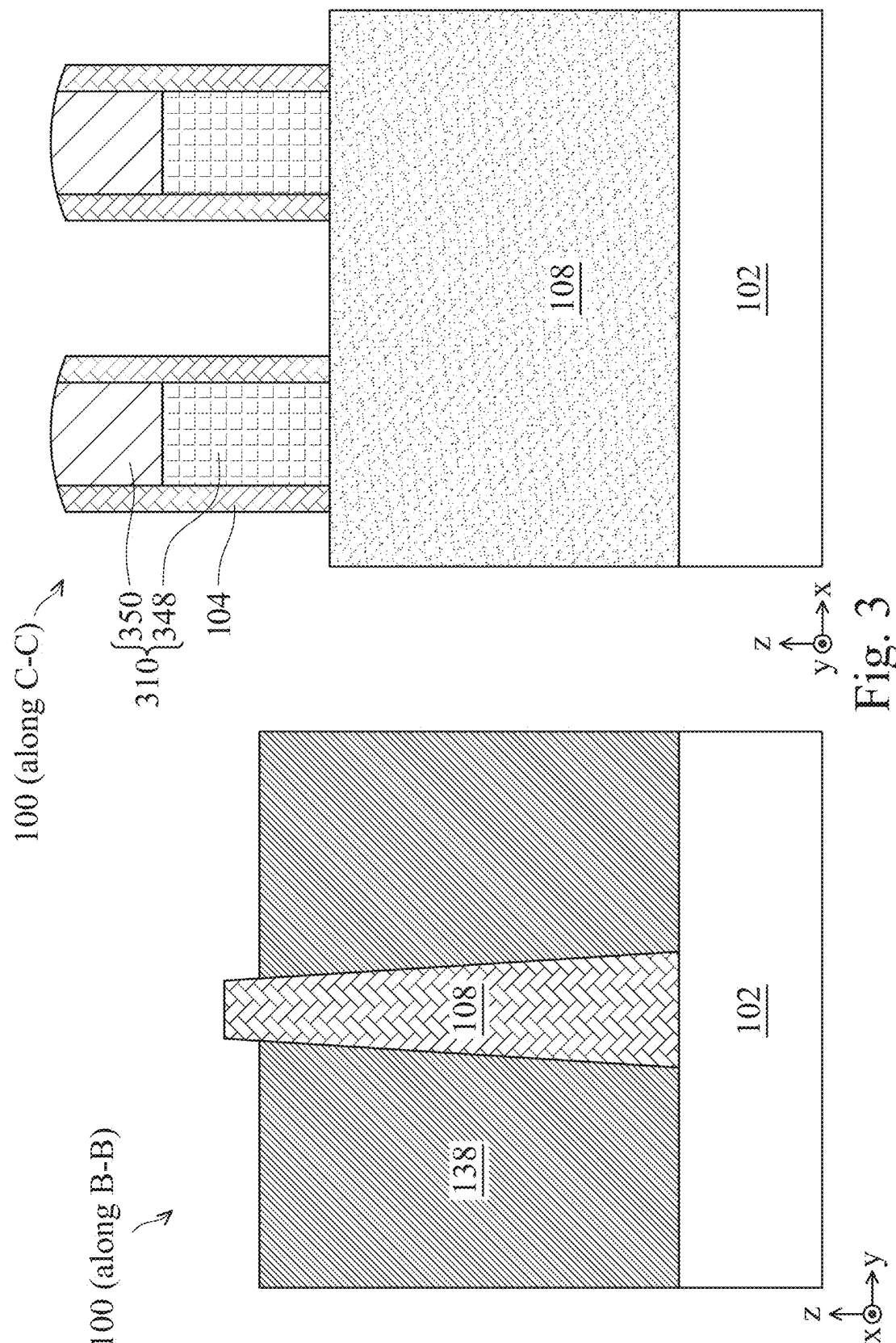


Fig. 3

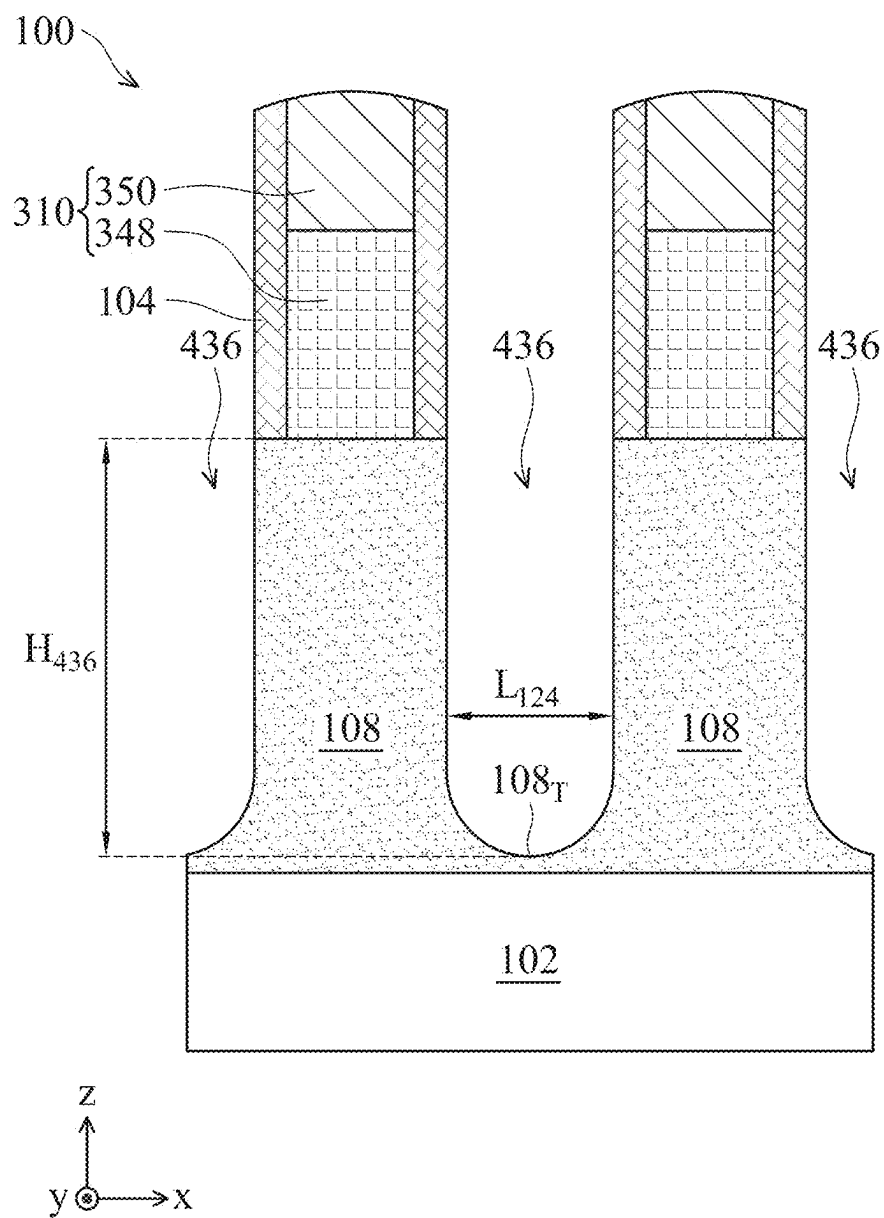


Fig. 4

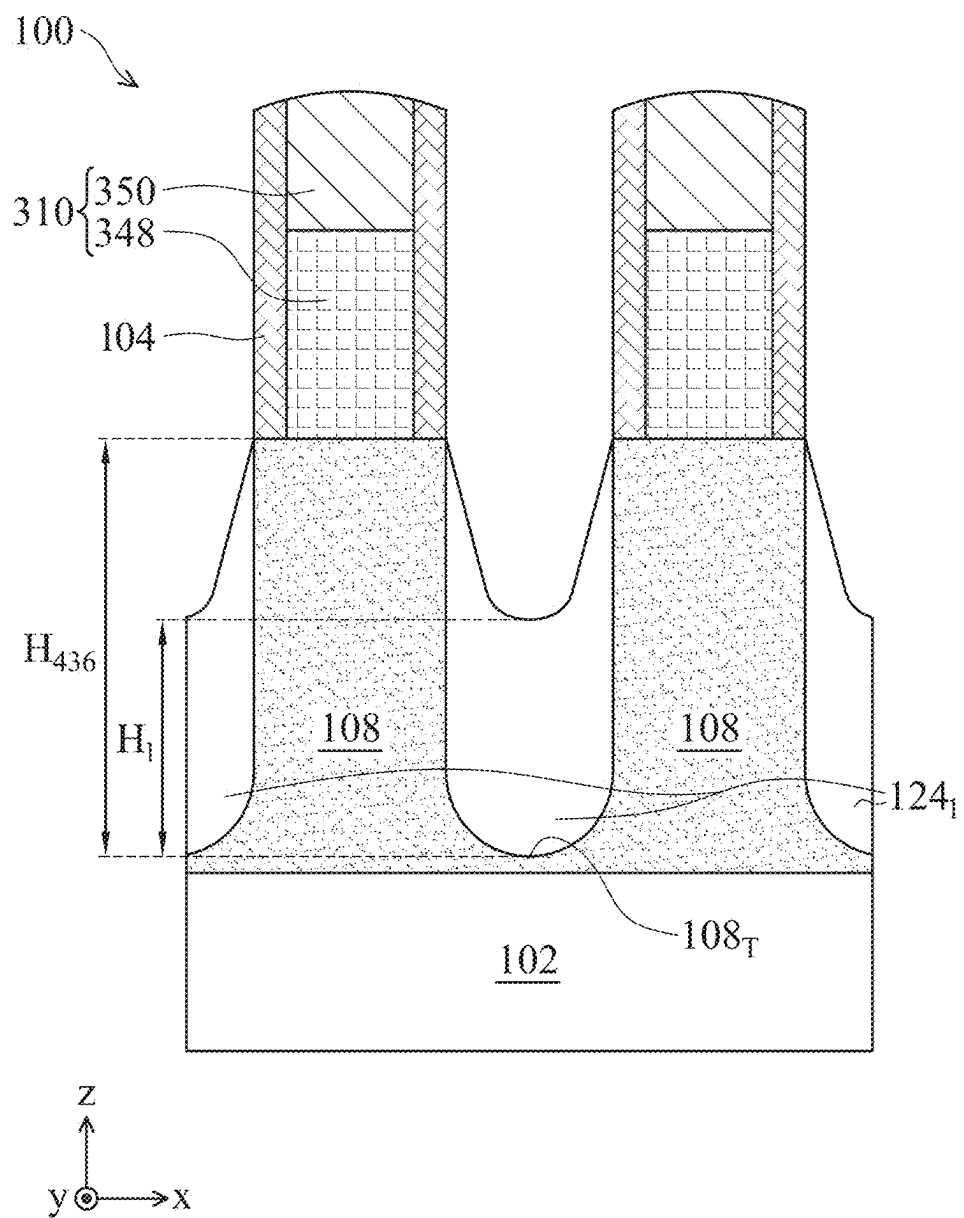


Fig. 5

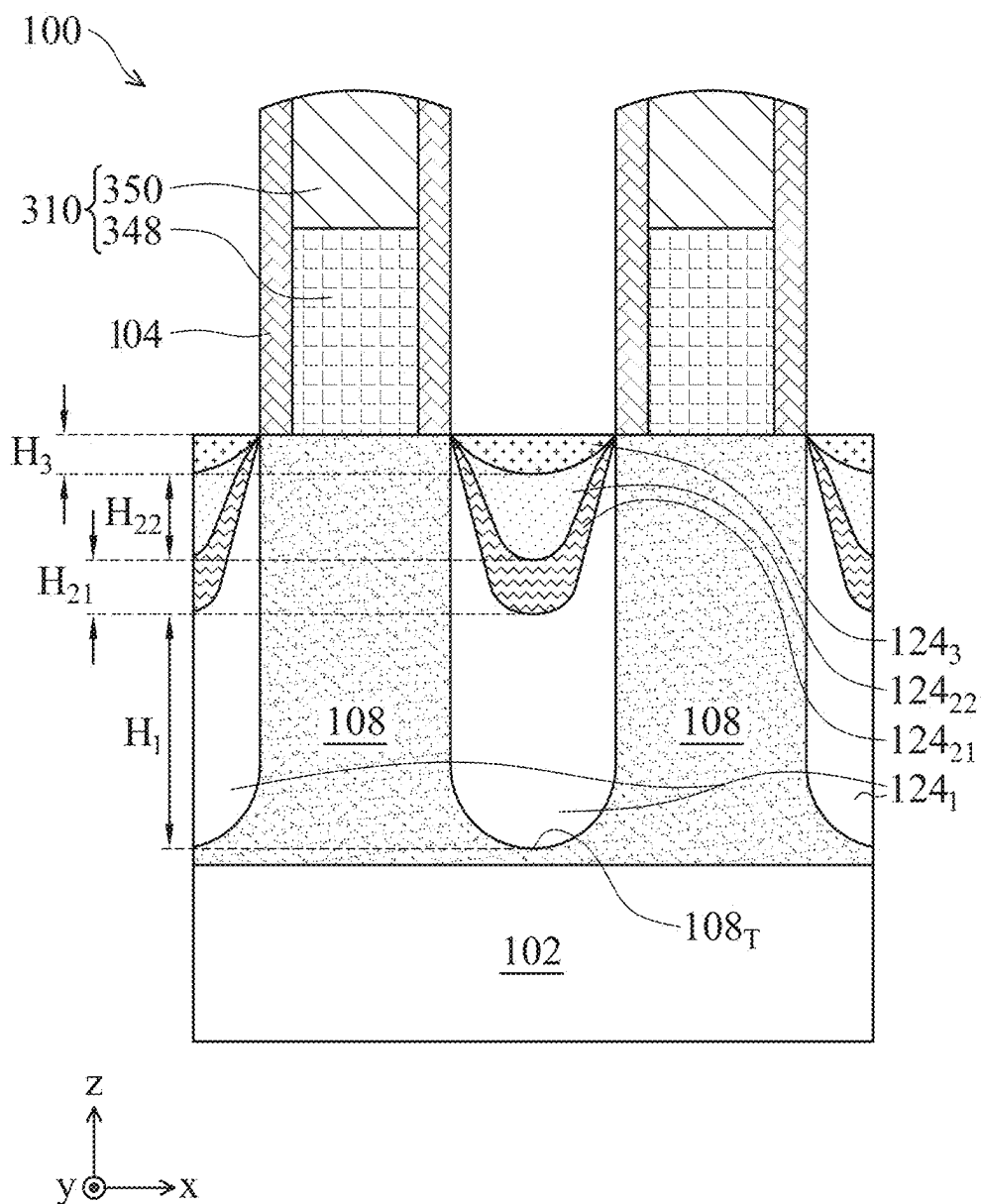


Fig. 6

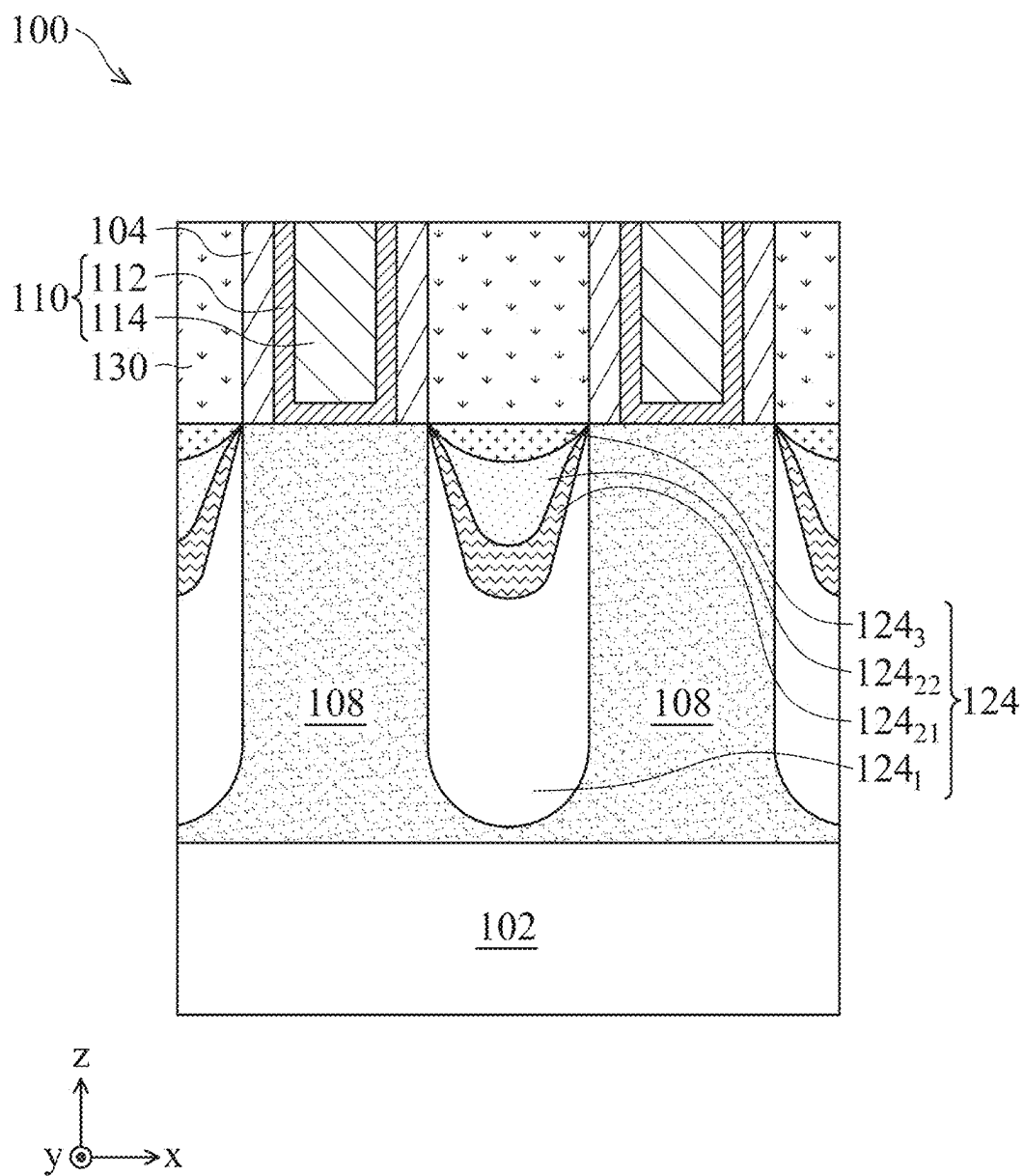


Fig. 7

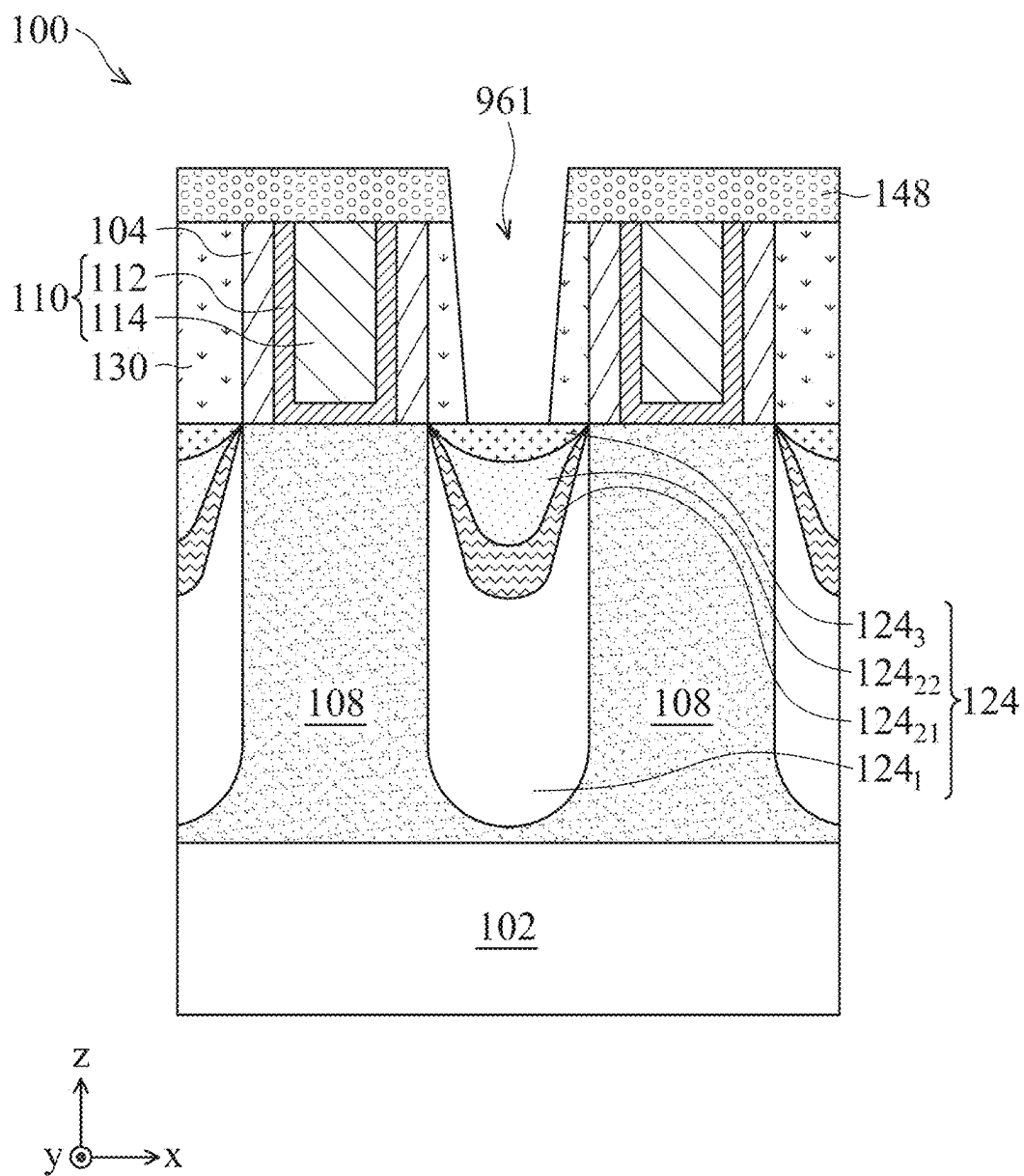


Fig. 8

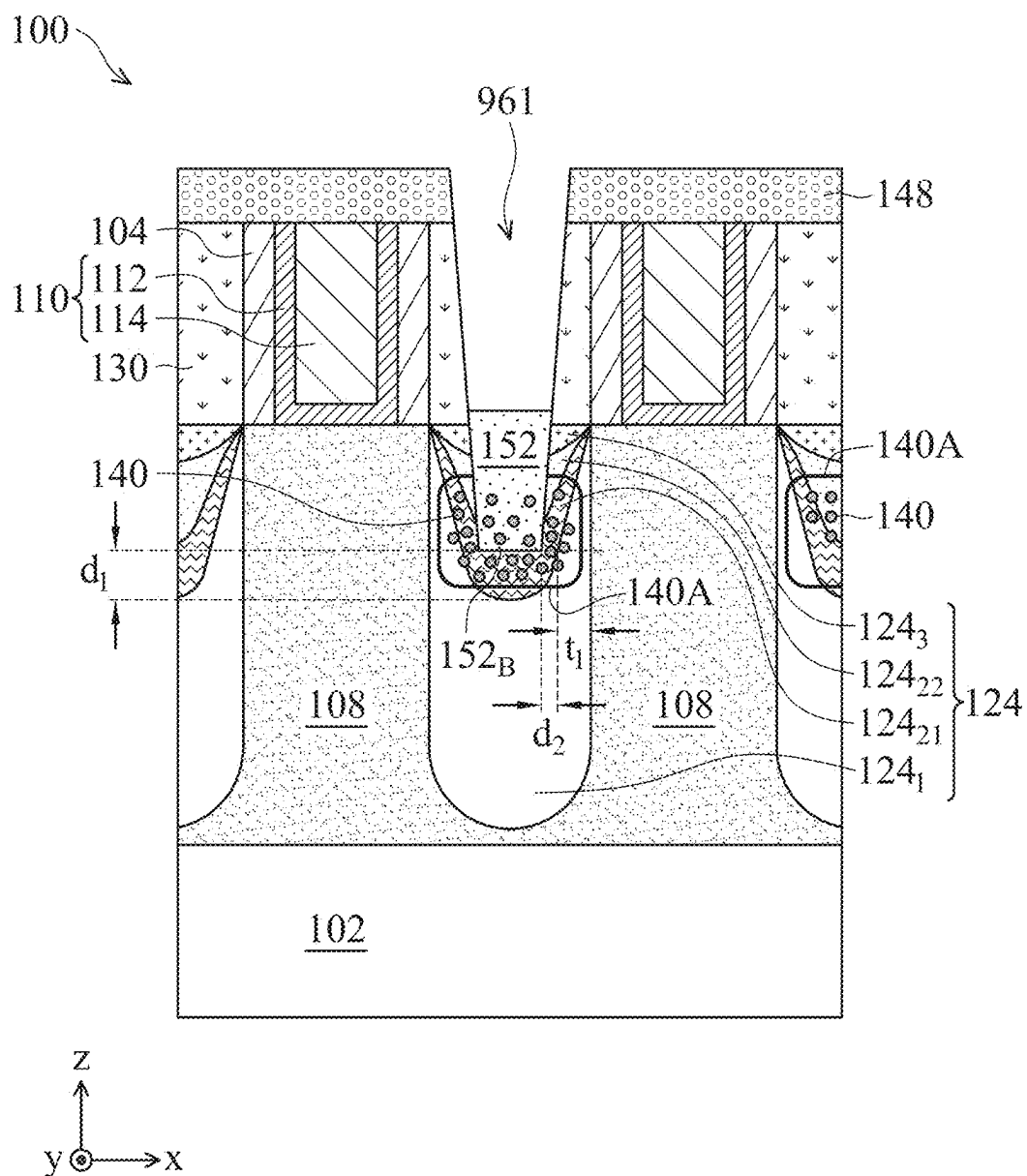


Fig. 9

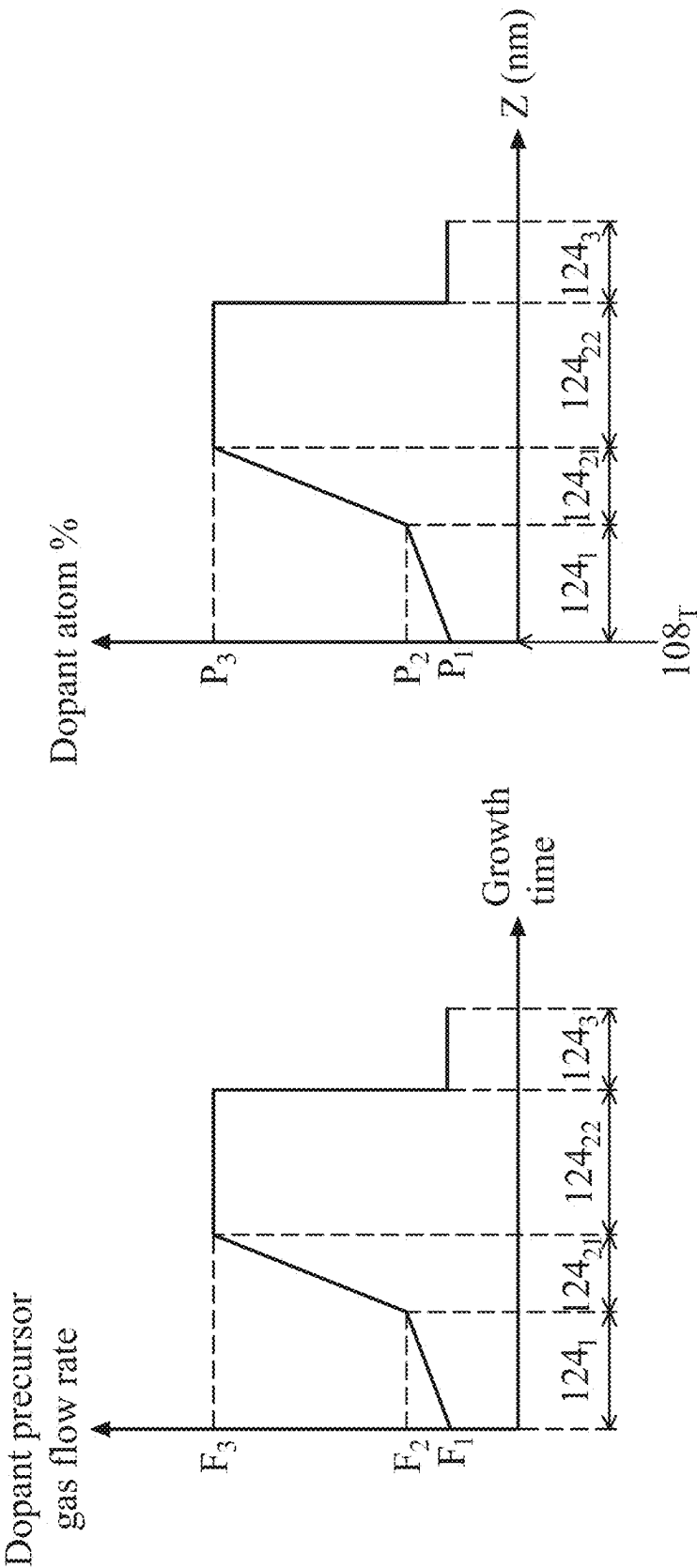


Fig. 10

SOURCE/DRAIN STRUCTURE FOR SEMICONDUCTOR DEVICE

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a continuation of U.S. Non-provisional patent application Ser. No. 18/165,117, titled “Source/Drain Structure for Semiconductor Device,” filed on Feb. 6, 2023, which is a continuation of U.S. Non-provisional patent application Ser. No. 17/207,359, titled “Source/Drain Structure for Semiconductor Device,” filed on Mar. 19, 2021, now U.S. Pat. No. 11,575,026, issued on Feb. 7, 2023, both of which are incorporated herein by reference in their entireties.

BACKGROUND

[0002] Advances in semiconductor technology has increased the demand for transistors with higher performance for faster processing systems. To meet this demand, it is important to reduce the transistor’s contact resistance to minimize the transistor delay (e.g., RC delay). Heavily doped source/drain terminals can reduce the transistor’s contact resistance, thus increasing transistor speed.

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of this disclosure are best understood from the following detailed description when read with the accompanying figures.

[0004] FIG. 1A is an isometric view of a semiconductor device, according to some embodiments.

[0005] FIGS. 1B and 1C are cross-sectional views of a semiconductor device, according to some embodiments.

[0006] FIG. 2 is a flow diagram of a method for fabricating a semiconductor device, according to some embodiments.

[0007] FIG. 3 illustrates cross-sectional views along lines B-B and C-C of structure of FIG. 1A at various stages of its fabrication, according to some embodiments.

[0008] FIGS. 4-9 are cross-sectional views of a semiconductor device at various stages of its fabrication process, according to some embodiments.

[0009] FIG. 10 illustrates a dopant precursor gas flow of a semiconductor device at various stages of its fabrication process and doping concentration profiles of a semiconductor device, according to some embodiments.

[0010] Illustrative embodiments will now be described with reference to the accompanying drawings. In the drawings, like reference numerals generally indicate identical, functionally similar, and/or structurally similar elements.

DETAILED DESCRIPTION

[0011] It is noted that references in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” “exemplary,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases do not necessarily refer to the same embodiment. Further, when a particular feature, structure or characteristic is described in connection with an embodiment, it would be within the knowledge of one skilled in the art to effect such feature, structure or characteristic in connection with other embodiments whether or not explicitly described.

[0012] It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by those skilled in relevant art(s) in light of the teachings herein.

[0013] Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0014] The term “nominal” as used herein refers to a desired, or target, value of a characteristic or parameter for a component or a process operation, set during the design phase of a product or a process, together with a range of values above and/or below the desired value. The range of values is typically due to slight variations in manufacturing processes or tolerances.

[0015] In some embodiments, the terms “about” and “substantially” can indicate a value of a given quantity that varies within 5% of the value (e.g., $\pm 1\%$, $\pm 2\%$, $\pm 3\%$, $\pm 4\%$, $\pm 5\%$ of the value). These values are merely examples and are not intended to be limiting. The terms “about” and “substantially” can refer to a percentage of the values as interpreted by those skilled in relevant art(s) in light of the teachings herein.

[0016] As used herein, the term “vertical” means nominally perpendicular to the surface of a substrate.

[0017] As used herein, the term “p-type” defines a structure, layer, and/or region as being doped with p-type dopants, such as boron.

[0018] As used herein, the term “n-type” defines a structure, layer, and/or region as being doped with n-type dopants, such as phosphorus.

[0019] Fins associated with fin field effect transistors (finFETs) or gate-all-around (GAA) FETs may be patterned by any suitable method. For example, the fins may be patterned using one or more photolithography processes, including a double-patterning process or a multi-patterning process. Double-patterning and multi-patterning processes can combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the fins.

[0020] A dopant is an impurity element that can be introduced to a semiconductor material to modulate the resistivity of the semiconductor material. For example, the dopant can be doped into the semiconductor material’s lattice structure as donors or acceptors to lower the semiconductor material’s resistivity. However, due to the dopant’s limited solid solubility in the semiconductor material, an amount of dopants that can be doped in the semiconductor material is limited, thus resulting in insufficient donors or acceptors in

the semiconductor material to effectively lower contact resistance in an integrated circuit (IC).

[0021] To address the aforementioned challenges, this disclosure is directed to a fabrication method and structures that provide clusters of a dopant embedded in source/drain (S/D) regions of a transistor. For example, the S/D regions can include a doped epitaxial stack. The epitaxial region can further include the clusters of the dopant to reduce the S/D contact resistance of the transistor. In some embodiments, the epitaxial stack can include a first epitaxial layer doped with the dopant of a first concentration, and a second epitaxial layer, formed over the first epitaxial layer, doped with the dopant of a second concentration greater than the first concentration. The second epitaxial layer can further include the clusters of the dopant to lower the S/D contact resistance. In some embodiments, the epitaxial stack can include multilayers of the silicon germanium (SiGe) layers. In some embodiments, the dopant can include boron; thus, clusters of boron can be formed embedded in the S/D region. A benefit of the present disclosure, among other, is to provide S/D regions with low contact resistance, thus improving transistor performance (e.g., speed).

[0022] A semiconductor device **100** having multiple FETs **101** formed over a substrate **102** is described with reference to FIGS. 1A-1C, according to some embodiments. FIG. 1A illustrates an isometric view of semiconductor device **100**, according to some embodiments. FIG. 1B illustrates a cross-sectional view along a source/drain (S/D) region (e.g., line B-B of FIG. 1A) of semiconductor device **100**, according to some embodiments. FIG. 1C illustrates a cross-sectional view along a channel direction (e.g., line C-C of FIG. 1A) of semiconductor device **100**, according to some embodiments. The discussion of elements in FIGS. 1A-1C with the same annotations applies to each other, unless mentioned otherwise. Semiconductor device **100** can be included in a microprocessor, memory cell, or other integrated circuit (IC). Also, even though each FET **101** shown in FIGS. 1A-1C is a fin field effect transistor (finFET), each FET **101** can be a gate-all-around (GAA) FET, according to some embodiments.

[0023] Referring to FIG. 1A, substrate **102** can be a semiconductor material, such as silicon. In some embodiments, substrate **102** can include a crystalline silicon substrate (e.g., wafer). In some embodiments, substrate **102** can include (i) an elementary semiconductor, such as silicon (Si) or germanium (Ge); (ii) a compound semiconductor including silicon carbide (SiC), gallium arsenide (GaAs), gallium phosphide (GaP), indium phosphide (InP), indium arsenide (InAs), and/or indium antimonide (InSb); (iii) an alloy semiconductor including silicon germanium carbide (Si-GeC), silicon germanium (SiGe), gallium arsenic phosphide (GaAsP), gallium indium phosphide (GaInP), gallium indium arsenide (GaInAs), gallium indium arsenic phosphide (GaInAsP), aluminum indium arsenide (AlInAs), and/or aluminum gallium arsenide (AlGaAs); or (iv) a combination thereof. Further, substrate **102** can be doped depending on design requirements (e.g., p-type substrate or n-type substrate). In some embodiments, substrate **102** can be doped with p-type dopants (e.g., boron (B), indium (In), aluminum (Al), or gallium (Ga)) or n-type dopants (e.g., phosphorus (P) or arsenic (As)).

[0024] Each FET **101** can include a fin structure **108** extending along an x-axis, a gate structure **110** traversing through fin structure **108** along an y-axis, and S/D regions

124 formed over portions of fin structure **108**. Although FIG. 1A shows each fin structure **108** accommodating two FETs **101**, any number of FETs **101** can be disposed along each fin structure **108**. In some embodiments, FET **101** can include multiple fin structures **108** extending along a first horizontal direction (e.g., in the x-direction) and gate structure **110** traversing through the multiple fin structures **108** along a second horizontal direction (e.g., in the y-direction). In some embodiments, multiple FETs **101** can have a common gate structure **110**.

[0025] Each fin structure **108** can be formed over substrate **102** and can include a material similar to substrate **102**. In some embodiments, fin structure **108** can include a material having a lattice constant substantially equal to (e.g., lattice mismatch within 5%) that of substrate **102**. In some embodiments, fin structure **108** can include a material identical to substrate **102**. In some embodiments, fin structure **108** can include multiple channel layers under gate structure **110**, each made of identical or different materials from each other. Fin structure **108** can be p-type doped, n-type doped, or un-doped. In some embodiments, a portion of fin structure **108** wrapped by gate structure **110** and another portion of fin structure **108** adjacent to gate structure **110** can be doped differently.

[0026] Semiconductor device **100** can further include shallow trench isolation (STI) regions **138** configured to provide electrical isolation between fin structures **108**. Also, STI regions **138** can provide electrical isolation between FETs **101** and neighboring active and passive elements (not shown in FIG. 1A) integrated with or deposited on substrate **102**. STI regions **138** can include one or more layers of dielectric material, such as a nitride layer, an oxide layer disposed on the nitride layer, and an insulating layer disposed on the nitride layer. In some embodiments, the insulating layer can refer to a layer that functions as an electrical insulator (e.g., a dielectric layer). In some embodiments, the insulating layer can include silicon oxide, silicon nitride, silicon oxynitride, fluorine-doped silicate glass (FSG), a low-k dielectric material, and/or other suitable insulating materials.

[0027] Referring to FIGS. 1A and 1C, gate structure **110** can be multi-layered structures that wraps around portions of one or more fin structures **108** to modulate FET **101**. In some embodiments, gate structure **110** can be referred to as gate-all-around (GAA) structures, where FET **101** can be referred to as a GAA FET **101**. Gate structure **110** can include a gate dielectric layer **112**, a gate electrode **114** disposed on dielectric layer **112**, and gate spacers **104** disposed on sidewalls of gate electrode **114**.

[0028] Gate dielectric layer **112** can be wrapped around portions of fin structure **108** and can be further disposed between gate electrode **114** and S/D regions **124** to prevent an electrical short in between. Gate dielectric layer **112** can include any suitable dielectric material with any suitable thickness that can provide channel modulation for FET **101**. In some embodiments, gate dielectric layer **112** can include silicon oxide and a high-k dielectric material (e.g., hafnium oxide or aluminum oxide), and gate dielectric layer **112** can have a thickness ranging from about 1 nm to about 5 nm. Other materials and thicknesses for gate dielectric layer **112** are within the scope and spirit of this disclosure.

[0029] Gate electrode **114** can function as a gate terminal for FET **101**. Gate electrode **114** can include a metal stack wrapping around portions of fin structure **108**. Gate elec-

trode **114** can include any suitable conductive material that provides a suitable work function to modulate FET **101**. In some embodiments, gate electrode **114** can include titanium nitride, tantalum nitride, tungsten nitride, titanium, aluminum, copper, tungsten, tantalum, or nickel (Ni). Other materials for gate electrode **114** are within the scope and spirit of this disclosure.

[0030] Gate spacer **104** can be in physical contact with gate dielectric layers **112**, according to some embodiments. Gate spacer **104** can have a low-k material with a dielectric constant less than about 3.9. For example, gate spacer **104** can include insulating material, such as silicon oxide, silicon nitride, a low-k material, and a combination thereof. In some embodiments, gate spacer **104** can have a thickness ranging from about 2 nm to about 10 nm. Other materials and thicknesses for gate spacer **104** are within the scope and spirit of this disclosure.

[0031] Each FET **101** can have S/D regions **124** formed at opposite sides (e.g., along x-direction) of gate structure **110**. Each S/D region **124** can include an epitaxially-grown semiconductor material that embeds dopant clusters **140**. In some embodiments, S/D region **124** can further include dopant clusters **140** embedded in the epitaxially-grown semiconductor material. In some embodiments, the epitaxially grown semiconductor material can be the same material as the material of substrate **102**. In some embodiments, the epitaxially-grown semiconductor material can have a lattice constant substantially equal to (e.g., lattice mismatch within 5%) that of the material of substrate **102**. In some embodiments, the epitaxially-grown semiconductor material can include (i) a semiconductor material, such as Ge and Si; (ii) a compound semiconductor material, such as GaAs and AlGaAs; or (iii) a semiconductor alloy, such as SiGe and GaAsP. The epitaxially-grown semiconductor material of S/D region **124** can be doped with p-type dopants, n-type dopants, or intrinsic dopants. In some embodiments, the doped p-type dopants, the doped n-type dopants, or the doped intrinsic dopants can be respectively incorporated to occupy S/D region **124**'s lattice sites, where each of the doped p-type dopants, the doped n-type dopants, or the doped intrinsic dopants can be proximate to S/D region **124**'s atoms. In some embodiments, the p-type dopants can include boron, indium, aluminum, gallium, zinc, beryllium, or magnesium. In some embodiments, the n-type dopants can include phosphorus, arsenic, sulfur, or selenium. In some embodiments, the intrinsic dopants can include iron or chromium. In some embodiments, S/D regions **124** formed at different FETs **101** can have different materials and/or doping types from each other.

[0032] Referring to FIGS. 1B and 1C, S/D region **124** can be a layer stack of epitaxially-grown semiconductor material. For example, S/D region can include a first layer **124₁** formed over fin structure **108**. For example, first layer **124₁** can be formed over a top surface **108_T** of portions of fin structure **108** that are laterally (e.g., in the x-direction) outside gate structure **110**. First layer **124₁** can have a vertical thickness H_1 measured from top surface **108_T**. In some embodiments, vertical thickness H_1 can be from about 8 nm to about 40 nm, from 12 nm to about 40 nm, from about 12 nm to about 30 nm, or from about 12 nm to about 16 nm to form dopant clusters **140** in S/D region **124**, as achieved by method **200** (discussed below). If vertical thickness H_1 is below the above-noted lower limits, the separation between silicide layer **152** (discussed below) and

first layer **124₁** can be increased to degrade the driving force for forming dopant clusters **140** in S/D region **124** (discussed below in method **200**). If vertical thickness H_1 is beyond the above-noted upper limits, the volume of second layer **124₂₁** (discussed below) can be reduced to degrade the stress for enhancing FET **101**'s channel mobility. In some embodiments, vertical thickness H_1 can be greater than about 8 nm, greater than about 15 nm, greater than about 20 nm, greater than about 25 nm, greater than about 30 nm, or greater than about 40 nm to form dopant clusters **140** in S/D region **124**, as achieved by method **200** (discussed below). If vertical thickness H_1 is below the above-noted lower limits, the separation between silicide layer **152** (discussed below) and first layer **124₁** can be increased to degrade the driving force for forming dopant clusters **140** in S/D region **124** (discussed below in method **200**). In some embodiments, S/D region **124** can have a length L_{124} (e.g., in the x-direction) parallel to fin structure **108**, where a ratio of vertical thickness H_1 to length L_{124} can be from about 0.1 to about 0.9, such as from about 0.1 to about 0.7, from about 0.1 to about 0.6, from about 0.3 to about 0.6, and from about 0.3 to about 0.5, to form dopant clusters **140** in S/D region **124**. If the ratio of vertical thickness H_1 to length L_{124} is below the above-noted lower limits, the separation between silicide layer **152** (discussed below) and first layer **124₁** can be increased to degrade the driving force for forming dopant clusters **140** in S/D region **124** (discussed below in method **200**). If the ratio of vertical thickness H_1 to length L_{124} is beyond the above-noted upper limits, the volume of second layer **124₂₁** (discussed below) can be reduced to degrade the stress for enhancing FET **101**'s channel mobility. In some embodiments, first layer **124₁** can be made of a compound semiconductor, such as SiGe. In some embodiments, first layer **124₁** can be made of SiGe with the atomic concentration of germanium less than about 35% to form dopant clusters **140** in S/D region **124**. If the atomic concentration of germanium of first layer **124₁** is beyond the above-noted upper limit, first layer **124₁** may not provide sufficient host material (e.g., silicon) to form silicide **152** proximate to first layer **124₁**, thus degrading the driving force for forming dopant clusters **140** in S/D region **124** (discussed below in method **200**). In some embodiments, first layer **124₁** can be made of SiGe with the atomic concentration of germanium from about 10% to about 35%, from about 10% to about 25%, or from about 10% to about 15% to form dopant clusters **140** in S/D region **124**. If the atomic concentration of germanium of first layer **124₁** is beyond the above-noted upper limits, first layer **124₁** may not provide sufficient host material (e.g., silicon) to form silicide **152** proximate to first layer **124₁**, thus degrading the driving force for forming dopant clusters **140** in S/D region **124** (discussed below in method **200**). If the atomic concentration of germanium of first layer **124₁** is below the above-noted lower limits, first layer **124₁** may not reconcile lattice mismatches between second layer **124₂₁** and substrate **102**, thus forming crystal defects (e.g., dislocations) in S/D region **124**. In some embodiments, first layer **124₁** can include SiGe with the atomic concentration of germanium gradually increasing along a vertical direction (e.g., in the z-direction) away from top surface **108_T**.

[0033] In some embodiments, first layer **124₁** can be doped with p-type dopants, such as boron, indium, aluminum, gallium, zinc, beryllium, and magnesium. In some embodiments, first layer **124₁** can be doped with p-type

dopants having a concentration less than $5 \times 10^{20}/\text{cm}^3$, such as from about $1 \times 10^{18}/\text{cm}^3$ to about $5 \times 10^{20}/\text{cm}^3$, from about $1 \times 10^{19}/\text{cm}^3$ to about $5 \times 10^{20}/\text{cm}^3$ or from about $1 \times 10^{19}/\text{cm}^3$ to about $3 \times 10^{20}/\text{cm}^3$. If first layer **124₁**'s dopant concentration is beyond the above-noted upper limit, FET **101** can be susceptible to the short channel effect. If first layer **124₁**'s dopant concentration is below the above-noted lower limits, FET **101** may have a higher contact resistance. In some embodiments, first layer **124₁** can be doped with p-type dopants having a concentration gradually increasing along a vertical direction (e.g., in the z-direction) away from top surface **108_T**. In some embodiments, first layer **124₁** can be doped with n-type dopants, such as phosphorus and arsenic, or intrinsic dopants, such as iron and chromium.

[0034] S/D region **124** can further include a second layer **124₂₁** formed over first layer **124₁**. In some embodiments, second layer **124₂₁** can contact first layer **124₁** along a junction **J₁**. Second layer **124₂₁** can have a vertical thickness **H₂₁** measured from portions of junction **J₁** that are laterally (e.g., in the x-y plane) in the perimeter of top surface **108_T**. In some embodiments, a ratio of vertical thickness **H₂₁** to length **L₁₂₄** can be less than about 0.8, less than about 0.7, less than about 0.5, less than about 0.3, or less than about 0.2 to form dopant clusters **140** in S/D region **124**, otherwise the separation between silicide layer **152** (discussed below) and the underlying first layer **124₁** can be increased to degrade the driving force for forming dopant clusters **140** in S/D region **124** (discussed below in method **200**). In some embodiments, a ratio of vertical thickness **H₂₁** to first layer **124₁**'s vertical thickness **H₁** can be less than about 20, less than about 15, less than about 10, less than about 5, less than about 3, or less than about 1 to form dopant clusters **140** in S/D region **124**. If the ratio of vertical thickness **H₂₁** to length **L₁₂₄** is beyond the above-noted upper limits, the separation between silicide layer **152** (discussed below) and the underlying first layer **124₁** can be increased to degrade the driving force for forming dopant clusters **140** in S/D region **124** (discussed below in method **200**). Second layer **124₂₁** can be made of a compound semiconductor, such as SiGe. In some embodiments, first and second layers **124₁** and **124₂₁** can be made of SiGe, where second layer **124₂₁** can have a greater atomic concentration of germanium than first layer **124₁** to provide a sufficient stress to enhance FET **101**'s channel mobility. For example, second layer **124₂₁** can be made of SiGe with the atomic concentration of germanium from about 35% to about 60%. In some embodiments, second layer **124₂₁** can include SiGe with the atomic concentration of germanium gradually increasing along a vertical direction (e.g., in the z-direction) away from junction **J₁**.

[0035] In some embodiments, second layer **124₂₁** can be doped with p-type dopants, such as boron, indium, aluminum, gallium, zinc, beryllium, and magnesium. Second layer **124₂₁** can be doped having a greater concentration than first layer **124₁** to lower FET **101**'s contact resistance. For example, first and second layers **124₁** and **124₂₁** can be made of SiGe doped with boron, where the boron concentration doped in second layer **124₂₁** can be greater than that in first layer **124₁**. In some embodiments, second layer **124₂₁** can be doped with p-type dopants having a concentration close to the solid solubility of the p-type dopants in second layer **124₂₁**, thus providing a condition that favors forming dopant clusters **140** of the p-type dopants in S/D region **124**, as achieved by method **200** (discussed below). For example,

boron's solid solubility in SiGe can be substantially equal to or greater than about $1 \times 10^{21}/\text{cm}^3$. Accordingly, second layer **124₂₁** can be made of SiGe doped with boron having a concentration substantially equal to or greater than about $1 \times 10^{21}/\text{cm}^3$, such as greater than about $2 \times 10^{21}/\text{cm}^3$, greater than about $3 \times 10^{21}/\text{cm}^3$, greater than about $3.5 \times 10^{21}/\text{cm}^3$, and greater than about $4 \times 10^{21}/\text{cm}^3$, to form dopant clusters **140** of boron. If second layer **124₂₁** is doped with the p-type dopant (e.g., boron) having a concentration below the above-noted lower limits, the driving force provided by silicide layer **152** may not precipitate the p-type dopant (e.g., boron) doped in second layer **124₂₁** to form dopant clusters **140** in S/D region **124**. In some embodiments, second layer **124₂₁** can be made of SiGe doped with boron having a concentration from about $5 \times 10^{20}/\text{cm}^3$ to about $6 \times 10^{21}/\text{cm}^3$, from about $8 \times 10^{20}/\text{cm}^3$ to about $6 \times 10^{21}/\text{cm}^3$, from about $1 \times 10^{21}/\text{cm}^3$ to about $6 \times 10^{21}/\text{cm}^3$, from about $2 \times 10^{21}/\text{cm}^3$ to about $6 \times 10^{21}/\text{cm}^3$, from about $3 \times 10^{21}/\text{cm}^3$ to about $6 \times 10^{21}/\text{cm}^3$, or from about $4 \times 10^{21}/\text{cm}^3$ to about $6 \times 10^{21}/\text{cm}^3$ to form dopant clusters **140** of boron achieved by method **200** (discussed below). If second layer **124₂₁** is doped with the p-type dopant (e.g., boron) having a concentration below the above-noted lower limits, the driving force provided by silicide layer **152** may not precipitate the p-type dopant (e.g., boron) doped in second layer **124₂₁** to form dopant clusters **140** in S/D region **124**. Further, the above-noted upper limit of the p-type dopant (e.g., boron) concentration may be limited or determined by the p-type dopant's solid solubility in second layer **124₂₁**, according to some embodiments. In some embodiments, second layer **124₂₁** can be doped with p-type dopants having a concentration gradually increasing along a vertical direction (e.g., in the z-direction) away from junction **J₁**. In some embodiments, second layer **124₂₁** can be doped with n-type dopants, such as phosphorus and arsenic, or intrinsic dopants, such as iron and chromium.

[0036] S/D region **124** can further include a third layer **124₂₂** formed over second layer **124₂₁**. In some embodiments, third layer **124₂₂** can contact second layer **124₂₁** along a junction **J₂**. Third layer **124₂₂** can have a vertical thickness **H₂₂** measured from portions of junction **J₂** that are laterally (e.g., in the x-y plane) in the perimeter of top surface **108_T**. In some embodiments, a ratio of vertical thickness **H₂₂** to length **L₁₂₄** can be less than about 0.5, less than about 0.3, less than about 0.1, less than about 0.08, less than about 0.06, or less than about 0.05 to form dopant clusters **140** achieved by method **200** (discussed below). If the ratio of vertical thickness **H₂₂** to length **L₁₂₄** is beyond the above-noted upper limits, the separation between silicide layer **152** (discussed below) and the underlying first layer **124₁** can be increased to degrade the driving force for forming dopant clusters **140** in S/D region **124** (discussed below in method **200**). In some embodiments, a ratio of vertical thickness **H₁** to vertical thickness **H₂₂** can be greater than about 0.5, greater than about 1, greater than about 2, greater than about 2.5, greater than about 3, or greater than about 4 to form dopant clusters **140** in S/D region **124**. If the ratio of vertical thickness **H₁** to vertical thickness **H₂₂** is below the above-noted lower limits, first layer **124₁** may not provide sufficient host material (e.g., silicon) to form silicide **152** proximate to first layer **124₁**, thus degrading the driving force for forming dopant clusters **140** in S/D region **124** (discussed below in method **200**). Third layer **124₂₂** can be made of a compound semiconductor, such as SiGe. In some embodiments, second and third layers **124₂₁** and **124₂₂** can

be made of SiGe, where third layer **124**₂₂ can have an atomic concentration of germanium greater than or substantially equal to second layer **124**₂₁ to provide sufficient stress to enhance FET **101**'s channel mobility. For example, second layer **124**₂₁ can be made of SiGe with the atomic concentration of germanium from about 35% to about 60%, and third layer **124**₂₂ can be made of SiGe with the atomic concentration of germanium of about 60%. In some embodiments, third layer **124**₂₂ can include SiGe with the atomic concentration of germanium substantially constant along a vertical direction (e.g., in the z-direction) away from junction **J**₂. In some embodiments, second and third layer **124**₂₁ and **124**₂₂ can include SiGe with the atomic concentration of germanium gradually increasing along a vertical direction (e.g., in the z-direction) away from junction **J**₂, where third layer **124**₂₂ can have a narrower range of the atomic concentration of germanium than second layer **124**₂₁. For example, second layer **124**₂₁ can include SiGe with the atomic concentration of germanium increasing from about 40% proximate to junction **J**₁ to about 55% away from junction **J**₁, where third layer **124**₂₂ can include SiGe with the atomic concentration of germanium increasing from about 55% proximate to junction **J**₂ to about 60% away from junction **J**₂.

[0037] In some embodiments, third layer **124**₂₂ can be doped with p-type dopants, such as boron, indium, aluminum, gallium, zinc, beryllium, and magnesium. Third layer **124**₂₂ can be doped having a substantially equal or a greater concentration than second layer **124**₂₁. For example, second and third layers **124**₂₁ and **124**₂₂ can be made of SiGe doped with boron, where the boron concentration doped in third layer **124**₂₂ can be a substantially equal to or a greater than that in second layer **124**₂₁ to lower FET **101**'s contact resistance. In some embodiments, third layer **124**₂₂ can be doped with p-type dopants having a concentration close to the solid solubility of the p-type dopants in third layer **124**₂₂, thus providing a condition that favors forming dopant clusters **140** of the p-type dopants in S/D region **124**, as achieved by method **200** (discussed below). For example, as previously discussed, boron's solid solubility in SiGe can be substantially equal to or greater than about $1 \times 10^{21}/\text{cm}^3$. Accordingly, third layer **124**₂₂ can be made of SiGe doped with boron having a concentration substantially equal to or greater than about $1 \times 10^{21}/\text{cm}^3$, such as greater than about $2 \times 10^{21}/\text{cm}^3$, greater than about $3 \times 10^{21}/\text{cm}^3$, greater than about $3.5 \times 10^{21}/\text{cm}^3$, and greater than about $4 \times 10^{21}/\text{cm}^3$, to form dopant clusters **140** of boron in S/D region **124**. If third layer **124**₂₂ is doped with the p-type dopant (e.g., boron) having a concentration below the above-noted lower limits, the driving force provided by silicide layer **152** may not precipitate the p-type dopant (e.g., boron) doped in third layer **124**₂₂ to form dopant clusters **140** in S/D region **124**. In some embodiments, third layer **124**₂₂ can be doped with boron having a concentration from about $5 \times 10^{20}/\text{cm}^3$ to about $6 \times 10^{21}/\text{cm}^3$, from about $8 \times 10^{20}/\text{cm}^3$ to about $6 \times 10^{21}/\text{cm}^3$, from about $1 \times 10^{21}/\text{cm}^3$ to about $6 \times 10^{21}/\text{cm}^3$, from about $2 \times 10^{21}/\text{cm}^3$ to about $6 \times 10^{21}/\text{cm}^3$, from about $3 \times 10^{21}/\text{cm}^3$ to about $6 \times 10^{21}/\text{cm}^3$, or from about $4 \times 10^{21}/\text{cm}^3$ to about $6 \times 10^{21}/\text{cm}^3$ to form dopant clusters **140** of boron in S/D region **124**. If third layer **124**₂₂ is doped with the p-type dopant (e.g., boron) having a concentration below the above-noted lower limits, the driving force provided by silicide layer **152** may not precipitate the p-type dopant (e.g., boron) doped in third layer **124**₂₂ to form dopant clusters **140**

in S/D region **124**. The above-noted upper limit of p-type dopant (e.g., boron) concentration may be limited or determined by the p-type dopant's solid solubility in third layer **124**₂₂, according to some embodiments. In some embodiments, third layer **124**₂₂ can be doped with p-type dopants having a concentration gradually increasing along a vertical direction (e.g., in the z-direction) away from junction **J**₂. In some embodiments, third layer **124**₂₂ can be doped with p-type dopants having a concentration substantially constant along a vertical direction (e.g., in the z-direction) away from junction **J**₂. In some embodiments, third layer **124**₂₂ can be doped with n-type dopants, such as phosphorus and arsenic, or intrinsic dopants, such as iron and chromium.

[0038] S/D region **124** can further include a capping layer **124**₃ formed over third layer **124**₂₂. In some embodiments, capping layer **124**₃ can contact third layer **124**₂₂ along a junction **J**₃. Capping layer **124**₃ can have a vertical thickness **H**₃ measured from portions of junction **J**₃ that are laterally (e.g., in the x-y plane) in the perimeter of top surface **108**_T. In some embodiments, vertical thickness **H**₃ can be from greater than about 2 nm, such as greater than about 5 nm, greater than about 10 nm, greater than about 20 nm, and greater than about 30 nm, to form dopant clusters **140** in S/D region **124**, as achieved by method **200** (discussed below). If vertical thickness **H**₃ is below the above-noted lower limits, capping layer **124**₃ may not provide sufficient host material (e.g., silicon) to form silicide **152** proximate to the underlying first layer **124**₁, thus degrading the driving force for forming dopant clusters **140** in S/D region **124** (discussed below in method **200**). In some embodiments, a ratio of vertical thickness **H**₃ to length **L**₁₂₄ can be greater than about 0.02, greater than about 0.04, greater than about 0.06, greater than about 0.1, or greater than about 0.15 to form dopant clusters **140** in S/D region **124**. If the ratio of vertical thickness **H**₃ to length **L**₁₂₄ is below the above-noted lower limits, capping layer **124**₃ may not provide sufficient host material (e.g., silicon) to form silicide **152** proximate to the underlying first layer **124**₁, thus degrading the driving force for forming dopant clusters **140** in S/D region **124** (discussed below in method **200**). In some embodiments, a ratio of vertical thickness **H**₃ to vertical thickness **H**₂₁ can be greater than about 0.03, greater than about 0.05, greater than about 0.1, greater than about 0.15, or greater than about 0.2 to form dopant clusters **140** in S/D region **124**. If the ratio of vertical thickness **H**₃ to vertical thickness **H**₂₁ is below the above-noted lower limits, capping layer **124**₃ may not provide sufficient host material (e.g., silicon) to form silicide **152** proximate to the underlying first layer **124**₁, thus degrading the driving force for forming dopant clusters **140** in S/D region **124** (discussed below in method **200**). In some embodiments, a ratio of vertical thickness **H**₃ to vertical thickness **H**₂₂ can be greater than about 0.03, greater than about 0.05, greater than about 0.1, greater than about 0.2, greater than about 0.3, or greater than about 0.4 to form dopant clusters **140** in S/D region **124**. If the ratio of vertical thickness **H**₃ to vertical thickness **H**₂₂ is below the above-noted lower limits, capping layer **124**₃ may not provide sufficient host material (e.g., silicon) to form silicide **152** proximate to the underlying first layer **124**₁, thus degrading the driving force for forming dopant clusters **140** in S/D region **124** (discussed below in method **200**). In some embodiments, capping layer **124**₃ can be made of a compound semiconductor, such as SiGe. In some embodiments, capping layer **124**₃ can be made of SiGe with the atomic

concentration of germanium less than about 50%, less than about 40%, less than about 30%, or less than about 25% to form dopant clusters 140 in S/D region 124, as achieved by method 200 (discussed below). If the atomic concentration of germanium of capping layer 124₃ is beyond the above-noted upper limits, capping layer 124₃ may not provide sufficient host material (e.g., silicon) to form silicide 152 proximate to first layer 124₁, thus degrading the driving force for forming dopant clusters 140 in S/D region 124 (discussed below in method 200). In some embodiments, capping layer 124₃ can be made of SiGe with the atomic concentration of germanium from about 5% to about 50%, from about 5% to about 40%, from about 5% to about 30%, from about 5% to about 25%, from about 5% to about 20%, or from about 5% to about 15% to form dopant clusters 140 in S/D region 124, as achieved by method 200 (discussed below). If the atomic concentration of germanium of capping layer 124₃ is beyond the above-noted upper limits, capping layer 124₃ may not provide sufficient host material (e.g., silicon) to form silicide 152 proximate to first layer 124₁, thus degrading the driving force for forming dopant clusters 140 in S/D region 124 (discussed below in method 200). If the atomic concentration of germanium of capping layer 124₃ is below the above-noted lower limits, the lattice mismatch between capping layer 124₃ and third layer 124₂₂ may be increased to cause crystal defects (e.g., dislocations) in S/D region 124. In some embodiments, capping layer 124₃ can include SiGe with the atomic concentration of germanium gradually increasing along a vertical direction (e.g., in the z-direction) away from junction J₃. In some embodiments, capping layer 124₃ can be made of pure silicon.

[0039] In some embodiments, capping layer 124₃ can be doped with p-type dopants, such as boron, indium, aluminum, gallium, zinc, beryllium, and magnesium. In some embodiments, capping layer 124₃ can be doped with p-type dopants having any suitable concentration lower than that of first layer 124₁, second layer 124₂₁, or third layer 124₂₂, such as having a concentration less than $3 \times 10^{20}/\text{cm}^3$. In some embodiments, capping layer 124₃ can be doped with p-type dopants having a concentration substantially constant along a vertical direction (e.g., in the z-direction) away from top surface 108₇. In some embodiments, capping layer 124₃ can be doped with n-type dopants, such as phosphorus and arsenic, or intrinsic dopants, such as iron and chromium.

[0040] Semiconductor device 100 can further include an interlayer dielectric (ILD) layer 130 to provide electrical isolation to structural elements it surrounds or covers, such as gate structure 110 and S/D regions 124. ILD layer 130 can include any suitable dielectric material to provide electrical insulation, such as silicon oxide, silicon dioxide, silicon oxycarbide, silicon oxynitride, silicon oxy-carbon nitride, and silicon carbonitride. ILD layer 130 can have any suitable thickness, such as from about 50 nm to about 200 nm, to provide electrical insulation. Based on the disclosure herein, other insulating materials and thicknesses for ILD layer 130 are within the scope and spirit of this disclosure.

[0041] Semiconductor device 100 can further include a layer of insulating material 148 formed over gate structure 110 and ILD layer 130. Layer of insulating material 148 can electrically isolate gate structure 110 and S/D region 124 from an interconnect structure (not shown in FIGS. 1A-1C) formed over FET 101. Layer of insulating material 148 can be made of any suitable insulating material, such as silicon

oxide, silicon nitride, a low-k dielectric material, and a high-k dielectric material. Further, layer of insulating material 148 can be made of any suitable thickness, such as from about 10 nm to about 400 nm, that can provide sufficient electrical insulation between FETs 101 and the interconnect structure (not shown in FIGS. 1A-1C) formed over FETs 101. Based on the disclosure herein, other insulating materials and thicknesses for layer of insulating material 148 are within the scope and spirit of this disclosure.

[0042] Semiconductor device 100 can further include a trench conductor structure 150 formed through layer of insulating material 148. Trench conductor structure 150 can electrically connect FET 101's S/D region 124 to the interconnect structure (not shown in FIGS. 1A-1C) formed over FET 101. In some embodiments, trench conductor structure 150 can connect FET 101's gate structure 110 to the interconnect structure formed over FET 101. In some embodiments, trench conductor structure 150 can protrude into S/D region 124. For example, as shown in FIGS. 1B and 1C, trench conductor structure 150 can include a silicide layer 152 protruding S/D region 124 and a layer of conductive material 154 formed over silicide layer 152. Silicide layer 152 can include a metal silicide material to provide a low resistance interface between layer of conductive material 154 and S/D region 124. In some embodiments, silicide layer 152 can be formed through S/D region 124's capping layer 124₃ and protruding S/D region 124's third layer 124₂₂. In some embodiments, silicide layer 152 can be formed through S/D region 124's capping layer 124₃ and third layer 124₂₂, and protruding S/D region 124's second layer 124₂₁. Silicide layer 152 can be a metal silicide that can include titanium, cobalt, nickel, platinum, palladium, tungsten, molybdenum, tantalum, vanadium, chromium, silicon, or germanium. Layer of conductive material 154 can include any suitable conductive material that provide low resistance between silicide layer 152 and the interconnect structure (not shown in FIGS. 1A-1C) formed over FET 101. For example, layer of conductive material 154 can include a metallic material, such as copper, tungsten, aluminum, and cobalt. In some embodiments, layer of conductive material 154 can further include a stack of conductive materials (not shown in FIGS. 1A-1C), such as a conductive nitride material (e.g., titanium nitride or tantalum nitride), that can act as a diffusion barrier, an adhesion promotion layer, or a nucleation layer to embed the above-noted metallic materials in layer of insulating material 148. Based on the disclosure herein, other materials for silicide layer 152 and layer of conductive material 154 are within the scope and spirit of this disclosure.

[0043] Silicide layer 152 can be formed proximate to S/D region 124's first layer 124₁ to form dopant clusters 140 in S/D region 124, as achieved by method 200 (discussed below). For example, as shown in FIG. 1C, first layer 124₁ can include a compound semiconductor material (e.g., SiGe) with a rich atomic percentage of a host material (e.g., atomic percentage of silicon greater than about 60%). Silicide layer 152 can be formed proximate to first layer 124₁ to result in a sufficient driving force to draw the host material (e.g., silicon) from first layer 124₁ towards silicide layer 152. Such driving force can precipitate the dopants (e.g., boron) doped in S/D region 124 to form dopant clusters 140 in S/D region 124. In some embodiments, a vertical (e.g., in the z-direction) separation d₁ between silicide layer 152's bottom surface 152_B and first layer 124₁ can be less than about 10

nm, such as from about 1 nm to about 10 nm, from about 2 nm to about 9 nm, and from about 3 nm to about 8 nm, to provide the sufficient driving force to form dopant clusters **140** in S/D region **124**. If vertical separation d_1 is below the above-noted lower limits, FET **101** may have a high contact resistance. If vertical separation d_1 is beyond the above-noted upper limits, the driving force that draws first layer **124**₁'s host material (silicon) for forming dopant clusters **140** can be degraded. In some embodiments, a horizontal (e.g., in the x-direction) separation d_2 between a portion of silicide layer **152**'s sidewall proximate to silicide layer **152**'s bottom surface **152**_B and first layer **124**₁ can be less than about 10 nm, such as from about 1 nm to about 9 nm, from about 2 nm to about 8 nm, and from about 3 nm to about 7 nm, to provide the sufficient driving force to form dopant clusters **140** in S/D region **124**. If horizontal separation d_2 is below the above-noted lower limits, FET **101** may have a high contact resistance. If horizontal separation d_2 is beyond the above-noted upper limits, the driving force that draws first layer **124**₁'s host material (silicon) for forming dopant clusters **140** can be degraded. In some embodiments, silicide layer **152** can be proximate to first layer **124**₁ to provide the sufficient driving force to form dopant clusters **140** in S/D region **124**, where a ratio of separation d_2 to silicide layer **152**'s bottom width L_{152} can be less than about 0.5, such as less than about 0.4, less than about 0.3, and less than about 0.2. If the ratio of separation d_2 to width L_{152} is beyond the above-noted upper limits, the driving force that draws first layer **124**₁'s host material (silicon) for forming dopant clusters **140** can be degraded. In some embodiments, silicide layer **152** can be proximate to first layer **124**₁ to provide the sufficient driving force to form dopant clusters **140** in S/D region **124**, where a ratio of silicide layer **152**'s bottom width L_{152} to S/D region **124**'s length L_{124} can be from about 0.5 to about 1.0, such as from about 0.5 to about 0.9, from about 0.5 to about 0.8, from about 0.5 to about 0.7, and from about 0.5 to about 0.6. If the ratio of width L_{152} to length L_{124} is below the above-noted lower limits, the driving force that draws first layer **124**₁'s host material (silicon) for forming dopant clusters **140** can be degraded. If the ratio of width L_{12} to length L_{124} is beyond the above-noted upper limits, FET **101** may have a high contact resistance. In some embodiments, first layer **124**₁ can have a side thickness (e.g., lateral thickness t_1) greater than a threshold to provide sufficient host material (e.g., silicon) to silicide layer **152** to form dopant clusters **140** in S/D region **124**. For example, first layer **124**₁ can have a lateral (e.g., in the x-direction) thickness t_1 proximate to a portion of first layer **124**₁ that vertically (e.g., in the z-direction) aligns with silicide layer **152**'s bottom surface **152**_B, where thickness t_1 can be greater than about 5 nm, such as greater than about 10 nm, greater than about 20 nm, greater than about 25 nm, and greater than about 30 nm, to provide sufficient host material (e.g., silicon) to silicide layer **152** to form dopant clusters **140** in S/D region **124**. If thickness t_1 is below the above-noted lower limits, first layer **124**₁ may not provide sufficient host material (e.g., silicon) to form silicide **152** proximate to first layer **124**₁, thus degrading the driving force for forming dopant clusters **140** in S/D region **124** (discussed below in method **200**). In some embodiments, separation of d_2 can represent second layer **124**₂₁'s lateral thickness formed over silicide layer **152**'s side surface, where a ratio of thickness t_1 to separation d_2 can be from about 0.6 to about 1.4, from about 0.7 to about 1.3, or from

about 0.8 to about 1.2 to provide sufficient host material (e.g., silicon) to silicide layer **152** to form dopant clusters **140** in S/D region **124**. If the ratio of thickness t_1 to separation d_2 is below the above-noted lower limits, the driving force that draws first layer **124**₁'s host material (silicon) for forming dopant clusters **140** can be degraded. If the ratio of thickness t_1 to separation d_2 is beyond the above-noted upper limits, FET **101** may have a high contact resistance.

[0044] Dopant cluster **140** can be embedded in S/D region **124** in the form of atom/molecule aggregations. For example, dopant cluster **140** can include multiple dopants (not shown in FIGS. **1B** and **1C**) that can be proximate to one another. In some embodiments, dopant cluster **140** can further include a vacancy structure (not shown in FIGS. **1B** and **1C**) that can be embedded in the multiple dopants of dopant cluster **140**. In some embodiments, dopant cluster **140** embedded in S/D region **124** can be different from dopants doped in S/D region **124**. For example, dopants doped in S/D region **124** can individually occupy a lattice site of S/D region **124**'s crystalline structure, while dopant cluster **140** can be an atom/molecule aggregation scattered in S/D region **124** and not incorporated in S/D region **124**'s lattice structure. In some embodiments, dopant cluster **140** can be made of a same material as the dopants in S/D region **124**. For example, S/D region **124** can be doped with a p-type dopant (e.g., boron), where such p-type doped S/D region **124** can further include dopant clusters **140** made of the same material of the p-type dopant (e.g., dopant clusters **140** can be boron clusters.) In some embodiments, a concentration of dopant clusters **140** of a dopant material (e.g., boron) in S/D region **124** can be greater than a maximum doping concentration of the same dopant material (e.g., boron) doped in S/D region **124** determined by the solid solubility limit of the dopant material (e.g., boron) in S/D region **124**. In some embodiments, dopant clusters **140** of a dopant material can be embedded in S/D region **124** with an atomic percentage from about 1% to about 10%, from about 2% to about 10%, from about 3% to about 8%, or from about 3% to about 6%, where the above-noted atomic percentage ranges can be substantially equal to or greater than the solid solubility of the dopant material in S/D region **124**. In some embodiments, dopant clusters **140** of a dopant material can be embedded in S/D region **124** with a concentration from about $3 \times 10^{20}/\text{cm}^3$ to about $6 \times 10^{22}/\text{cm}^3$, from about $1 \times 10^{21}/\text{cm}^3$ to about $3 \times 10^{22}/\text{cm}^3$, or from about $1 \times 10^{21}/\text{cm}^3$ to about $1 \times 10^{22}/\text{cm}^3$, where the above-noted atomic percentage ranges can be substantially equal to or greater than the solid solubility of the dopant material in S/D region **124**. In some embodiments, a maximum atomic percentage of a dopant material in each of S/D region **124**'s sub-regions (e.g., first layer **124**₁, second layer **124**₂₁, or third layer **124**₂₂) can be from about 1% to about 10%, from about 3% to about 10%, from about 3% to about 6%, where the dopant material can be doped S/D region **124** as the dopants and/or precipitated in S/D region **124** as dopant clusters **140**. In some embodiments, a maximum concentration of a dopant material in each of S/D region **124**'s sub-regions (e.g., first layer **124**₁, second layer **124**₂₁, or third layer **124**₂₂) can be from about $3 \times 10^{20}/\text{cm}^3$ to about $6 \times 10^{22}/\text{cm}^3$, from about $1 \times 10^{21}/\text{cm}^3$ to about $3 \times 10^{22}/\text{cm}^3$, or from about $1 \times 10^{21}/\text{cm}^3$ to about $1 \times 10^{22}/\text{cm}^3$, where the dopant material can be doped S/D region **124** as the dopants and/or precipitated in S/D region **124** as dopant clusters **140**. In some embodiments, by

embedding dopant clusters **140** with the above-noted atomic ranges in S/D region **124**, the effective bandgap of S/D region **124** can be reduced, thus increasing the carrier concentration (e.g., hole concentration) in S/D region **124** to lower the contact resistance associated with FET **101**.

[0045] In some embodiments, dopant clusters **140** can be distributed in a cluster region **140A** proximate to silicide layer **152**. For example, as shown in FIG. 1B, cluster region **140A** can vertically (e.g., in the z-direction) span from silicide layer **152**'s sidewall with a vertical dimension S_{140Z} less than about 10 nm, less than about 8 nm, less than about 5 nm, or less than about 3 nm. If dimension S_{140Z} is greater than the above-noted upper limits, the separation between silicide layer **152** (discussed below) and first layer **124**, can be increased to degrade the driving force for forming dopant clusters **140** in S/D region **124** (discussed below in method **200**). In some embodiments, cluster region **140A** can have a vertical (e.g., in the z-direction) span H_{140} substantially equal to or less than vertical dimension S_{140Z} . Similarly, as shown in FIG. 1C, cluster region **140A** can laterally (e.g., in the x-direction) span from silicide layer **152**'s sidewall with a lateral dimension S_{140X} substantially equal to or less than the combination (e.g., thickness t_1 +separation d_2) of thickness t_1 and separation d_2 . In some embodiments, cluster region **140A** can laterally (e.g., in the x-direction) span with a dimension substantially equal to S/D region **124**' length L_{124} . In some embodiments, cluster region **140A** can overlap with junctions J_1 , J_2 , and/or J_3 . In some embodiments, cluster region **140A** can overlap with silicide layer **152** (e.g., dopant clusters **140** formed in silicide layer **152**).

[0046] FIG. 2 is a flow diagram of a method **200** for fabricating semiconductor device **100**, according to some embodiments. For illustrative purposes, the operations illustrated in FIG. 2 will be described with reference to the example fabrication process for fabricating semiconductor device **100** as illustrated in FIGS. 3-10. FIG. 3 illustrates cross-sectional views along lines B-B and C-C of structure of FIG. 1A at various stages of its fabrication, according to some embodiments. FIGS. 4-9 illustrates cross-sectional views along line C-C of structure of FIG. 1A at various stages of its fabrication, according to some embodiments. FIG. 10 illustrates the usage of dopant precursor and the atomic concentration of doping clusters in method **200**, according to some embodiments. Operations can be performed in a different order or not performed depending on specific applications. Method **200** may not produce a complete semiconductor device **100**. Accordingly, it is understood that additional processes can be provided before, during, and/or after method **200**, and that some other processes may be briefly described herein. Further, the discussion of elements in FIGS. 1A-1C and 3-10 with the same annotations applies to each other, unless mentioned otherwise.

[0047] Referring to FIG. 2, in operation **205**, a recess structure is formed adjacent to a sacrificial gate structure. For example, a recess structure **436** (shown in FIG. 4) can be formed over substrate **102** and adjacent to a sacrificial gate structure **310** with reference to FIGS. 3 and 4. The process of forming recess structure **436** can include (i) forming fin structures **108** (shown in FIG. 3) over substrate **102**; (ii) forming sacrificial gate structures **310** (shown in FIG. 3) over fin structures **108**; and (iii) removing fin structures **108** through sacrificial gate structures **310** to form recess structure **436**.

[0048] Referring to FIG. 3, the process of forming fin structures **108** can include (i) providing substrate **102**; (ii) etching substrate **102** through a patterned mask layer (not shown in FIG. 3) using an etch process; and (iii) forming STI region **138** over the etched substrate **102** using a deposition process and an etch back process. The etch process for etching substrate **102** can include a dry etch process or a wet etch process. In some embodiments, the dry etch process can include using any suitable etchant, such as an oxygen-containing gas, a fluorine-containing gas, a chlorine-containing gas, and a bromine-containing gas, and the wet etch process can include etching in any suitable wet etchant, such as diluted hydrofluoric acid, potassium hydroxide solution, ammonia, and nitric acid. In some embodiments, the deposition process for forming STI region **138** can include any suitable growth process, such as a physical vapor deposition (PVD) process, a chemical vapor deposition (CVD) process, a high-density-plasma (HDP) CVD process, a flowable CVD (FCVD) process, and an atomic layer deposition (ALD) process. In some embodiments, the etch back process for forming STI region **138** can include a dry etch process, a wet etch process, or a polishing process, such as chemical mechanical polishing (CMP) process. Based on the disclosure herein, other processes for forming fin structures **108** are within the spirit and scope of this disclosure.

[0049] The process of forming sacrificial gate structure **310** can include (i) blanket depositing a polysilicon layer **348** and a hard mask layer **350** over fin structures **108** using a suitable deposition process, such as a CVD process, a PVD process, and an ALD process; (ii) removing polysilicon layer **348** and hard mask layer **350** through a patterned mask layer (not shown in FIG. 3) using an etching process; and (iii) forming gate spacers **104** over sidewalls of polysilicon layer **348** using a suitable deposition process and an etch process. Based on the disclosure herein, other processes for forming sacrificial gate structures **310** are within the spirit and scope of this disclosure.

[0050] After forming sacrificial gate structure **310**, recess structure **436** can be formed by removing fin structures **108** through sacrificial gate structures **310** using an etching process. The etching process can include a dry etch process or a wet etch process. In some embodiments, the etching process can be a time-etch process. In some embodiments, the dry etch process can include using any suitable etchant, such as an oxygen-containing gas, a fluorine-containing gas, a chlorine-containing gas, and a bromine-containing gas, and the wet etch process can include etching in any suitable wet etchant, such as diluted hydrofluoric acid, potassium hydroxide solution, ammonia, and nitric acid. The resulting recess structure **436** can span with length L_{124} associated with two adjacent sacrificial gate structure **310**'s separation. Further, the resulting recess structure **436** can protrude in fin structure **108** with a depth H_{436} to define top surface **108₇** of portions of fin structure **108** that are laterally (e.g., in the x-direction) outside sacrificial gate structure **310**. In some embodiments, depth H_{436} can be any suitable dimension, such as from about 30 nm to about 70 nm and from about 40 nm to about 60 nm. Based on the disclosure herein, other lengths and depths of recess structures **436** are within the spirit and scope of this disclosure.

[0051] Referring to FIG. 2, in operation **210**, a first epitaxial region is formed in the recess structure. For example, first layer **124₁** (shown in FIG. 5) can be formed in recess structure **436** of FIG. 4 with reference to FIGS. 5 and 10.

The process of forming first layer 124_1 can include epitaxially growing a semiconductor material of vertical thickness H_1 measured from top surface 108_T using an epitaxial growth process. In some embodiments, the epitaxial growth process can selectively grow first layer 124_1 over recess structure 436 's side surfaces and bottom surface (e.g., top surface 108_T), while exposing sacrificial gate structures 310 . The epitaxial growth process can include (i) a CVD process, such as a low pressure CVD (LPCVD) process, a rapid thermal CVD (RTCVD) process, a metal-organic CVD (MOCVD) process, an atomic layer CVD (ALCVD) process, an ultrahigh vacuum CVD (UHV-CVD) process, and a reduced pressure CVD (RPCVD) process; (ii) a molecular beam epitaxy (MBE) processes; (iii) an epitaxial deposition/partial etch process, such as a cyclic deposition-etch (CDE) process; or (iv) a selective epitaxial growth (SEG) process. The epitaxial process can be conducted using suitable processing gases associated with the semiconductor material of first layer 124_1 . For example, first layer 124_1 can include SiGe, where the processing gases can include silane (SiH_4), disilane (Si_2H_6), dichlorosilane (DCS), germane (GeH_4), hydrogen (H_2), and nitrogen (N_2). In some embodiments, first layer 124_1 can include a compound semiconductor material, where at least one element of the compound semiconductor material can be the host material that can be supplied to silicide layer 152 (formed at operation 225) to assist precipitating dopant clusters 140 (formed at operation 225) in S/D region 124 . For example, first layer 124_1 can include SiGe having the atomic concentration of silicon greater than about 65% to assist precipitating dopant clusters 140 in S/D region 124 . In some embodiments, first layer 124_1 can have thickness H_1 greater than a threshold to provide reduced separations d_1 and d_2 (shown in FIGS. $1C$ and 9) to silicide layer 152 (formed at operation 225) to assist precipitating dopant clusters 140 in S/D region 124 . For example, a ratio of first layer 124_1 's thickness H_1 to recess structure 436 's depth H_{436} can be greater than about 0.1, greater than about 0.2, or greater than about 0.3 to assist precipitating dopant clusters 140 in S/D region 124 .

[0052] The process of forming first layer 124_1 can further include doping the grown semiconductor layer with dopants during the epitaxial growth process. For example, referring to FIG. 10 , a dopant precursor gas with a flow rate F_1 can be included in the epitaxial growth process to result in first layer 124_1 in-situ doped with dopants having an atomic percentage P_1 . In some embodiments, first layer 124_1 can be in-situ doped with boron having atomic percentage P_1 of about 0.5%, where the associated dopant precursor gas can include diborane (B_2H_6), boron difluoride (BF_2), or boron trifluoride (BF_3) with flow rate F_1 about 100 standard cubic centimeters per minute (sccm). In some embodiments, dopant precursor gas flow can gradually increase from flow rate F_1 to a flow rate F_2 to result in first layer 124_1 in-situ doped with dopants having an atomic percentage gradually increasing from atomic percentage P_1 to an atomic percentage P_2 along a vertical direction (e.g., in the z-direction) away from top surface 108_T . In some embodiments, flow rate F_2 can be less than about 200 sccm to result in atomic percentage of P_2 less than about 1% to mitigate short channel effects of FET 101 . In some embodiments, the dopant precursor gas can include phosphine (PH_3) and arsine (AsH_3) that can respectively in-situ dope phosphorus and arsenic in first layer 124_1 . Based on the disclosure herein, other epitaxial growth processes, processing gases, dopant precursor gases, and dopant

precursor gas flow rates for forming first layer 124_1 are within the spirit and scope of this disclosure.

[0053] Referring to FIG. 2 , in operation 215 , a second epitaxial region is formed over the first epitaxial region. For example, second layer 124_{21} (shown in FIG. 6) can be formed over first layer 124_1 of FIG. 5 with reference to FIGS. 6 and 10 . The process of forming second layer 124_{21} can include epitaxially growing a semiconductor material of vertical thickness H_{21} over first layer 124_1 using an epitaxial growth process similar to that for growing first layer 124_1 . In some embodiments, first and second layers 124_1 and 124_{21} can include SiGe, where the epitaxial growth process for growing second layer 124_{21} can include supplying higher flow rate of GeH_4 to result in second layer 124_{21} with higher atomic ratio of germanium than first layer 124_1 . In some embodiments, second layer 124_{21} 's vertical thickness H_{21} can be thinned for silicide layer 152 (formed in second layer 124_{21} at operation 225) proximate to first layer 124_1 to attract first layer 124_1 's host material (e.g., silicon) to precipitate dopant clusters 140 in S/D region 124 . The process of forming second layer 124_{21} can further include doping the grown semiconductor layer with dopants during the epitaxial growth process for second layer 124_{21} . In some embodiments, second layer 124_{21} can be in-situ doped with dopants (e.g., boron) having an atomic percentage P_3 close to the dopant's solid solubility (e.g., boron's solid solubility in SiGe) in second layer 124_{21} by including a dopant precursor gas (e.g., B_2H_6) with a flow rate F_3 greater than flow rate F_2 . For example, the dopant's solid solubility in second layer 124_{21} can be substantially equal to or greater than about $3 \times 10^{20}/\text{cm}^3$, where second layer 124_{21} can be in-situ doped with dopants (e.g., boron) having atomic percentage P_3 substantially equal to greater than about $3 \times 10^{20}/\text{cm}^3$, such as greater than about $8 \times 10^{20}/\text{cm}^3$. The atom percentage P_3 can allow silicide layer 152 to extract the dopants (e.g., boron) from second layer 124_{21} to precipitate dopant clusters 140 (e.g., boron clusters) in S/D region 124 . In some embodiments, a maximum of atomic percentage P_3 can be greater than about 1%, greater than about 2%, greater than about 3%, greater than about 4%, greater than about 5%, or greater than about 6% for the p-type dopant doped in SiGe. In some embodiments, the dopant precursor gas flow can gradually increase from flow rate F_2 to a flow rate F_3 to result in second layer 124_{21} in-situ doped with dopants having an atomic percentage gradually increasing from atomic percentage P_2 to atomic percentage P_3 along a vertical direction (e.g., in the z-direction) away from top surface 108_T . Based on the disclosure herein, other epitaxial growth processes, processing gases, dopant precursor gases, and dopant precursor gas flow rates for forming second layer 124_{21} are within the spirit and scope of this disclosure.

[0054] In some embodiments, referring to FIGS. 6 and 10 , operation 215 can further include forming third layer 124_{22} over second layer 124_{21} . The process of forming third layer 124_{22} can include epitaxially growing a semiconductor material of vertical thickness H_{22} over second layer 124_{21} using an epitaxial growth process similar to that for growing second layer 124_{21} . In some embodiments, third layer 124_{22} 's vertical thickness H_{22} can be thinned for silicide layer 152 (formed in second layer 124_{21} at operation 225) formed through third layer 124_{22} and proximate to first layer 124_1 to attract first layer 124_1 's host material (e.g., silicon) to precipitate dopant clusters 140 in S/D region 124 . The process of forming third layer 124_{22} can further include

doping the grown semiconductor layer with dopants during the epitaxial growth process for third layer **124₂₂**. In some embodiments, third layer **124₂₂** can be in-situ doped with dopants (e.g., boron) having atomic percentage P_3 close to the dopant's solid solubility (e.g., boron's solid solubility in SiGe) in third layer **124₂₂** by including a dopant precursor gas (e.g., B_2H_6) with flow rate F_3 . For example, the dopant's solid solubility in third layer **124₂₂** can be substantially equal to or greater than about $1 \times 10^{21}/\text{cm}^3$, where third layer **124₂₂** can be in-situ doped with dopants (e.g., boron) having atomic percentage P_3 substantially equal to greater than about $1 \times 10^{21}/\text{cm}^3$, such as greater than about $2 \times 10^{21}/\text{cm}^3$. The atom percentage P_3 can allow silicide layer **152** to extract the dopants (e.g., boron) from third layer **124₂₂** to precipitate dopant clusters **140** (e.g., boron clusters) in S/D region **124**. In some embodiments, the dopant precursor gas flow can be substantially constant at flow rate F_3 to result in third layer **124₂₂** uniformly in-situ doped with dopants having atomic percentage P_3 along a vertical direction (e.g., in the z-direction) away from top surface **108_T**. Based on the disclosure herein, other epitaxial growth processes, processing gases, dopant precursor gases, and dopant precursor gas flow rates for forming third layer **124₂₂** are within the spirit and scope of this disclosure.

[0055] In some embodiments, referring to FIGS. 6 and 10, operation **215** can further include forming capping layer **124₃** over third layer **124₂₂**. The process of forming capping layer **124₃** can include epitaxially growing a semiconductor material of vertical thickness H_3 over third layer **124₂₂** using an epitaxial growth process similar to that for growing third layer **124₂₂**. In some embodiments, capping layer **124₃**'s vertical thickness H_3 can be thickened to provide sufficient host material (e.g., silicon) to silicide layer **152** (formed at operation **225**) to assist precipitating dopant clusters **140** (formed at operation **225**) in S/D region **124**. The process of forming capping layer **124₃** can further include doping the grown semiconductor layer with dopants during the epitaxial growth process for capping layer **124₃**. In some embodiments, capping layer **124₃** can be in-situ doped with dopants (e.g., boron) having an atomic percentage less than atomic percentage P_2 by including a dopant precursor gas (e.g., B_2H_6) with a flow rate less than flow rate F_2 . Based on the disclosure herein, other epitaxial growth processes, processing gases, dopant precursor gases, and dopant precursor gases flow rate for forming capping layer **124₃** are within the spirit and scope of this disclosure.

[0056] Referring to FIG. 2, in operation **220**, the sacrificial gate structure is replaced with a metal gate structure. For example, sacrificial gate structure **310** can be replaced with gate structure **110** with reference to FIG. 7. The process of forming gate structure **110** can include (i) forming ILD layer **130** over second layer **124₂₁** using a suitable deposition process, such as a PVD process and a CVD process; (ii) removing hard mask layer **350** to coplanarize polysilicon layer **348** with ILD layer **130** using a polishing process, such as a CMP process; (iii) removing polysilicon layer **348** to form a recess (not shown in FIG. 7) to expose fin structures **108** using an etching process; and (iv) filling gate dielectric layer **112** and a gate electrode **114** in the recess using a suitable deposition process, such as ALD, CVD, and PVD. Based on the disclosure herein, other processes for forming gate structure **110** are within the spirit and scope of this disclosure.

[0057] Referring to FIG. 2, in operation **225**, a metal contact is formed adjacent to the metal gate structure and in the second epitaxial region. For example, trench conductor structure **150** (shown in FIGS. 1A-1C) can be formed adjacent to gate structure **110** and in second layer **124₂₁** with reference to FIGS. 8 and 9. The process of forming trench conductor structure **150** can include (i) blanket depositing layer of insulating material **148** over the structure of FIG. 8 (e.g., over gate structure **110**, and S/D region **124**) via a suitable deposition process, such as a CVD process, a PVD process, and an ALD process; (ii) forming a recess structure **961** (shown in FIG. 8) through layer of insulating material **148** and ILD layer **130** using a lithography process and an etching process; (iii) forming silicide layer **152** in recess structure **961**; and (iv) forming layer of conductive material **154** (shown in FIGS. 1B and 1C) over silicide layer **152** using a suitable deposition process (e.g., a CVD process, an ALD process, a PVD process, and an e-beam evaporation process) and a polishing process (e.g., a CMP process).

[0058] The process of forming silicide layer **152** can include (i) depositing a layer of metallic material, such as titanium, cobalt, nickel, tungsten, and any other suitable metallic material over recess structure **961** to contact S/D region **124**; (ii) performing an annealing process to react a portion of the deposited layer of metallic material with S/D region **124**; and (iii) removing an un-reacted portion of the deposited layer of metallic material using an etching process. During the annealing process, the deposited layer of metallic material can consume (e.g., react with) the S/D region **124**'s host material (e.g., silicon) from a nearby region, such as cluster region **140A**. The respective solid solubility (dissolved amount) of the dopants in S/D region **124** within cluster region **140A** can therefore be decreased, thus releasing the dopants (e.g., boron) from the crystalline structure of S/D region **124** to form dopant clusters **140** of the dopant (e.g., boron clusters). For example, silicide layer **152** can be formed in second layer **124₂₁** and/or in third layer **124₂₂** doped with boron with a concentration ranging from atomic percentage P_2 to atomic percentage P_3 . The annealing process for forming silicide layer **152** can extract the dopants (e.g., boron) from second layer **124₂₁** and/or from third layer **124₂₂** to result in scattered doping clusters **140** (e.g., boron clusters) proximate to silicide layer **152** (e.g., within cluster region **140A**). In some embodiments, the annealing process can draw the host material from portions of S/D region **124** that is beyond the range of cluster region **140A**, such as drawing silicon from first layer **124₁** towards cluster region **140A** and/or drawing silicon from capping layer **124₃** towards cluster region **140A**. The annealing process can create a driving force to extract the dopants from S/D region **124** in cluster region **140A** to precipitate dopant clusters **140** of the dopants. Therefore, reducing separation d_1 , reducing separation d_2 , increasing bottom width L_{152} , and increasing thickness t_1 can enhance the precipitation of dopant clusters **140**, as discussed in FIG. 1C. In some embodiments, the annealing process can include an annealing temperature from about 700°C . to about 1000°C . to form silicide layer **152** associated with separation d_1 , separation d_2 , bottom width L_{152} , and/or thickness t_1 for forming dopant clusters **140**. In some embodiments, the annealing process can include an annealing time from about 0.001 seconds to about 0.1 seconds to form silicide layer **152** associated with separation d_1 , separation d_2 , bottom width L_{152} , and/or thickness t_1 for forming dopant clusters **140**. In some

embodiments, the annealing process can include a rapid thermal annealing (RTA) process or a spike thermal process using LASER to form silicide layer **152** associated with separation d_1 , separation d_2 , bottom width L_{152} , and/or thickness t_1 for forming dopant clusters **140**. Based on the disclosure herein, other annealing durations, annealing temperatures, or annealing methods for forming silicide layer **152** are within the spirit and scope of this disclosure.

[0059] The present disclosures provides an exemplary transistor S/D structure and a method for forming the same. The S/D structure can be included with cluster dopants to lower its bandgap to provide a low transistor contact resistance. In some embodiments, the S/D structure can include a first epitaxial layer and a second epitaxial layer formed over the first epitaxial layer. The first and second epitaxial layers can be doped with dopants. In some embodiments, the second epitaxial layer can further include dopant clusters of the dopant to lower the bandgap of the second epitaxial layer. In some embodiments, the method for forming the dopant clusters can include forming a silicide layer in the second epitaxial layer. The silicide layer can extract the dopants from the second epitaxial layer to precipitate the dopant clusters. A benefit of the S/D structures, among others, is to reduce the transistor contact resistance, thus improving transistor performance.

[0060] In some embodiments, a method can include providing a substrate, forming a first epitaxial layer over the substrate, and forming a second epitaxial layer over the first epitaxial layer. The process of forming the first epitaxial layer can include doping the first epitaxial layer with a first dopant having a first concentration. The process of forming the second epitaxial layer can include doping the second epitaxial layer with a second dopant having a second concentration greater than the first concentration. The method can further include precipitating the second dopant to form a cluster of the second dopant in the second epitaxial layer.

[0061] In some embodiments, a method can include forming a fin structure over a substrate, forming a gate structure over the fin structure, forming a recess structure in the fin structure and adjacent to the gate structure, forming a first epitaxial layer in the recess structure, and forming a second epitaxial layer over the first epitaxial layer. The process of forming the first epitaxial layer can include doping the first epitaxial layer with a first dopant having a first concentration. The process of forming the second epitaxial layer can include doping the second epitaxial layer with a second dopant having a second concentration greater than the first concentration. The method can further include precipitating the second dopant to form a cluster of the second dopant in the second epitaxial layer.

[0062] In some embodiments, a semiconductor structure can include a substrate, a fin structure over the substrate, a gate structure over the fin structure, an epitaxial region formed in the fin structure and adjacent to the gate structure. The epitaxial region can embed a plurality of clusters of dopants.

[0063] The foregoing disclosure outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art

should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A structure, comprising:

a fin structure on a substrate;

an epitaxial layer on the fin structure;

a silicide layer on the epitaxial layer; and

a plurality of dopant clusters in the epitaxial layer and the silicide layer.

2. The structure of claim 1, wherein the plurality of dopant clusters comprises dopants having a concentration greater than a maximum doping concentration determined by a solid solubility limit of the dopants in the epitaxial layer.

3. The structure of claim 1, wherein the epitaxial layer comprises a first layer and a second layer on the first layer, wherein a first gradient of dopant concentration of the first layer is less than a second gradient of dopant concentration of the second layer.

4. The structure of claim 3, wherein a distance between a first bottom surface of the silicide layer and a second bottom surface of the second layer is less than about 10 nm.

5. The structure of claim 3, wherein an interface between the silicide layer and the second layer is substantially flat.

6. The structure of claim 3, wherein a distance between a first side surface of the silicide layer and a second side surface of the second layer is less than about 10 nm.

7. The structure of claim 1, wherein the plurality of dopant clusters comprises boron.

8. A structure, comprising:

a channel region on a substrate; and

an source/drain (S/D) structure adjacent to the channel region, wherein the S/D structure comprises:

a first epitaxial layer in contact with the channel region;

a second epitaxial layer on the first epitaxial layer; and

a plurality of dopant clusters in the first and second epitaxial layers, wherein the plurality of dopant clusters comprises dopants, and wherein a first gradient of dopant concentration in the first epitaxial layer is less than a second gradient of dopant concentration in the second epitaxial layer.

9. The structure of claim 8, wherein the S/D structure further comprises a third epitaxial layer on the second epitaxial layer, and wherein a third gradient of dopant concentration in the third epitaxial layer is less than a second gradient of dopant concentration in the second epitaxial layer.

10. The structure of claim 8, wherein the S/D structure further comprises a third epitaxial layer on the second epitaxial layer, and wherein a dopant concentration of the third epitaxial layer is substantially constant.

11. The structure of claim 10, wherein the S/D structure further comprises a capping layer on the third epitaxial layer, and wherein a dopant concentration of the capping layer is less than the dopant concentration of the third epitaxial layer.

12. The structure of claim 10, wherein the dopant concentration of the third epitaxial layer is greater than a maximum doping concentration determined by a solid solubility limit of the dopants in the epitaxial layer.

13. The structure of claim 10, further comprising a silicide layer protruding through the third epitaxial layer and into the second epitaxial layer.

14. The structure of claim **13**, wherein the silicide layer comprises a portion of the plurality of dopant clusters.

15. A structure, comprising:

a fin structure on a substrate;

an epitaxial layer on the fin structure and comprising a cluster region doped with dopants, wherein a dopant concentration of the cluster region is greater than a maximum doping concentration determined by a solid solubility limit of the dopants in the epitaxial layer; and a silicide layer protruding into the cluster region.

16. The structure of claim **15**, wherein the cluster region comprises dopant clusters.

17. The structure of claim **15**, wherein the silicide layer comprises dopant clusters.

18. The structure of claim **15**, wherein the cluster region comprises a first layer in contact with the silicide layer and a second layer separated from the silicide layer by the first layer.

19. The structure of claim **18**, wherein a first gradient of dopant concentration of the first layer is greater than a second gradient of dopant concentration of the second layer.

20. The structure of claim **18**, wherein the cluster region further comprises a third layer on the second layer, and wherein a first dopant concentration at a first interface between the first and second layers is less than a second dopant concentration at a second interface between the third and second layers.

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